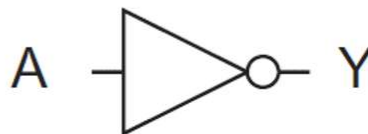
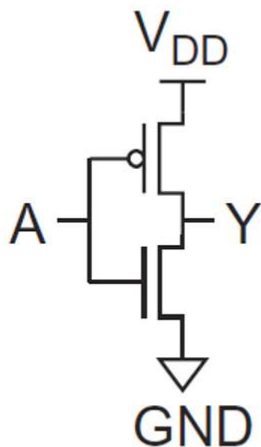


CMOS Inverter

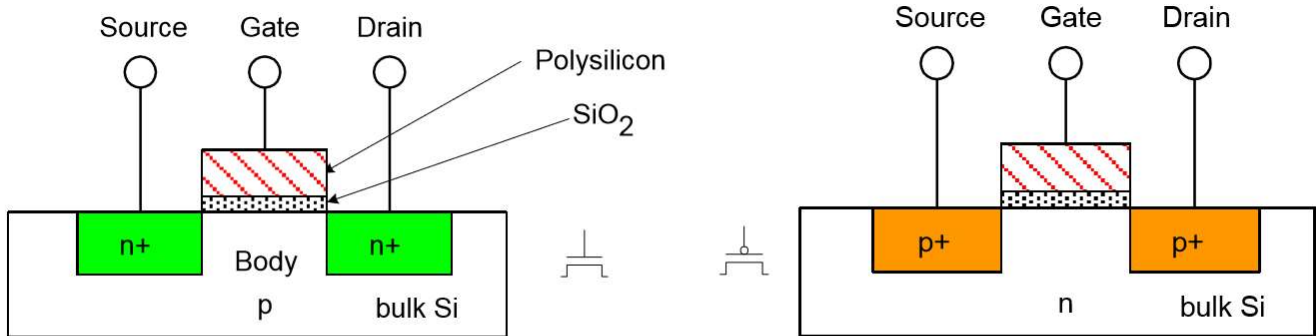
- ❑ Ideally, there is no static (idle) power consumption



A	Y

MOSFET

- ❑ MOSFET: Metal Oxide Semiconductor Field Effect Transistor
- ❑ Since 1970s: The gate has been formed from polycrystalline silicon (polysilicon)
- ❑ Metal gates reemerged in 2007 (45 nm technology) to solve materials problems in advanced manufacturing processes

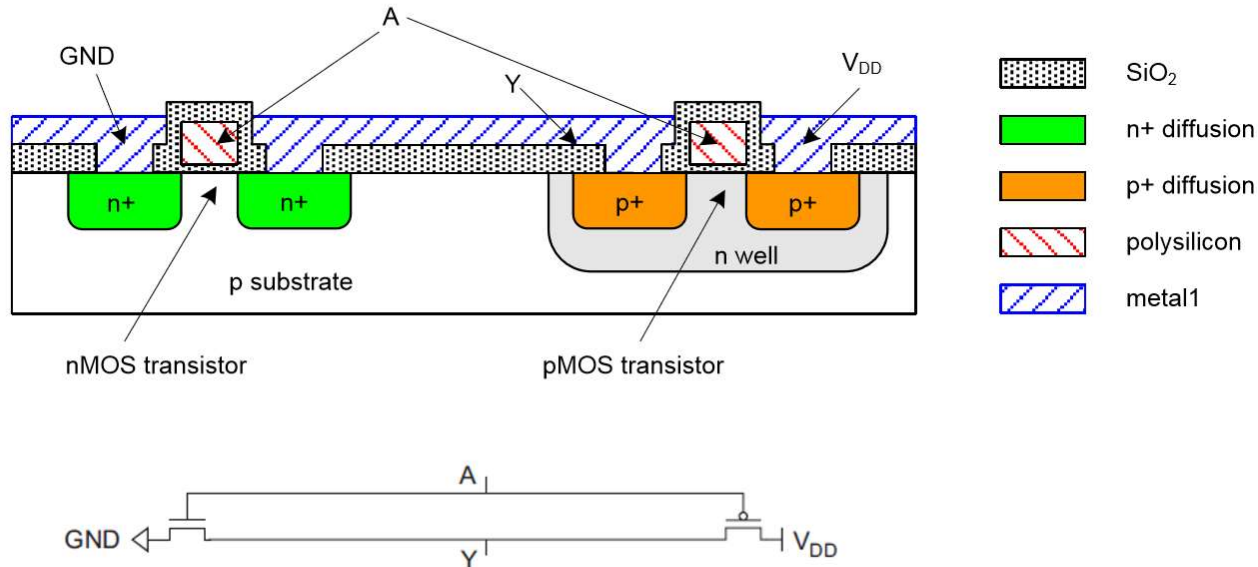


CMOS Fabrication

- ❑ CMOS: NMOS and PMOS transistors are fabricated on the same silicon wafer
- ❑ On each step, different materials are deposited or etched
- ❑ Lithography process similar to printing press
- ❑ Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process

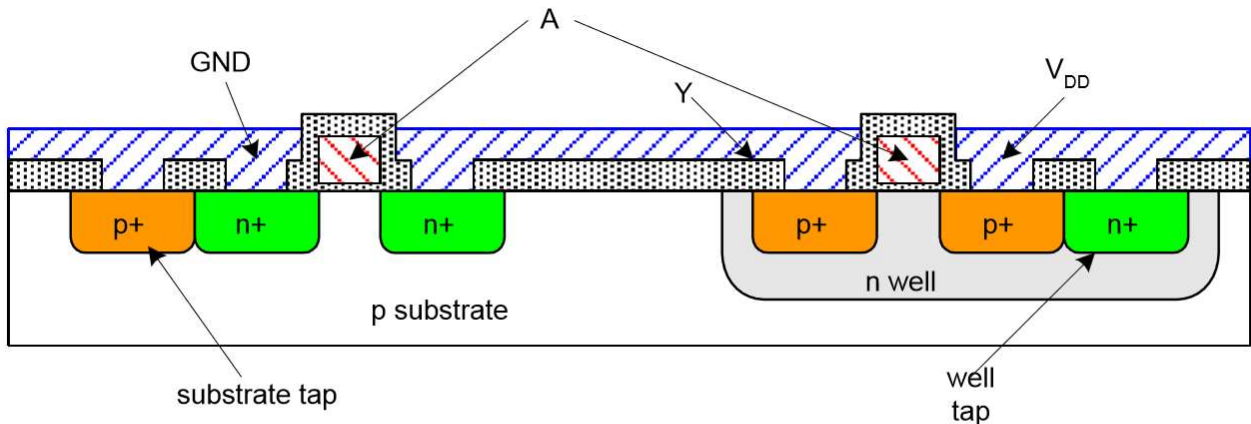
Inverter Cross-section

- ❑ Typically use p-type substrate for NMOS transistors
- ❑ Requires n-well for body of PMOS transistors



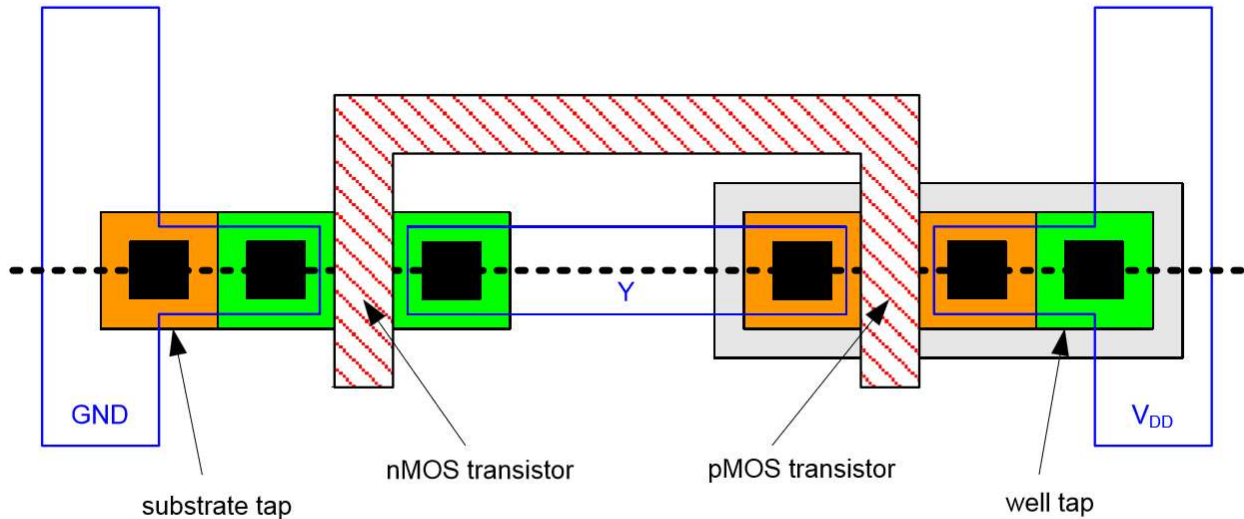
Well and Substrate Taps

- ❑ Substrate must be tied to GND and n-well to V_{DD}
- ❑ Metal to lightly-doped semiconductor forms a Schottky Diode
 - Use heavily doped well and substrate contacts / taps to create a low resistance ohmic contact

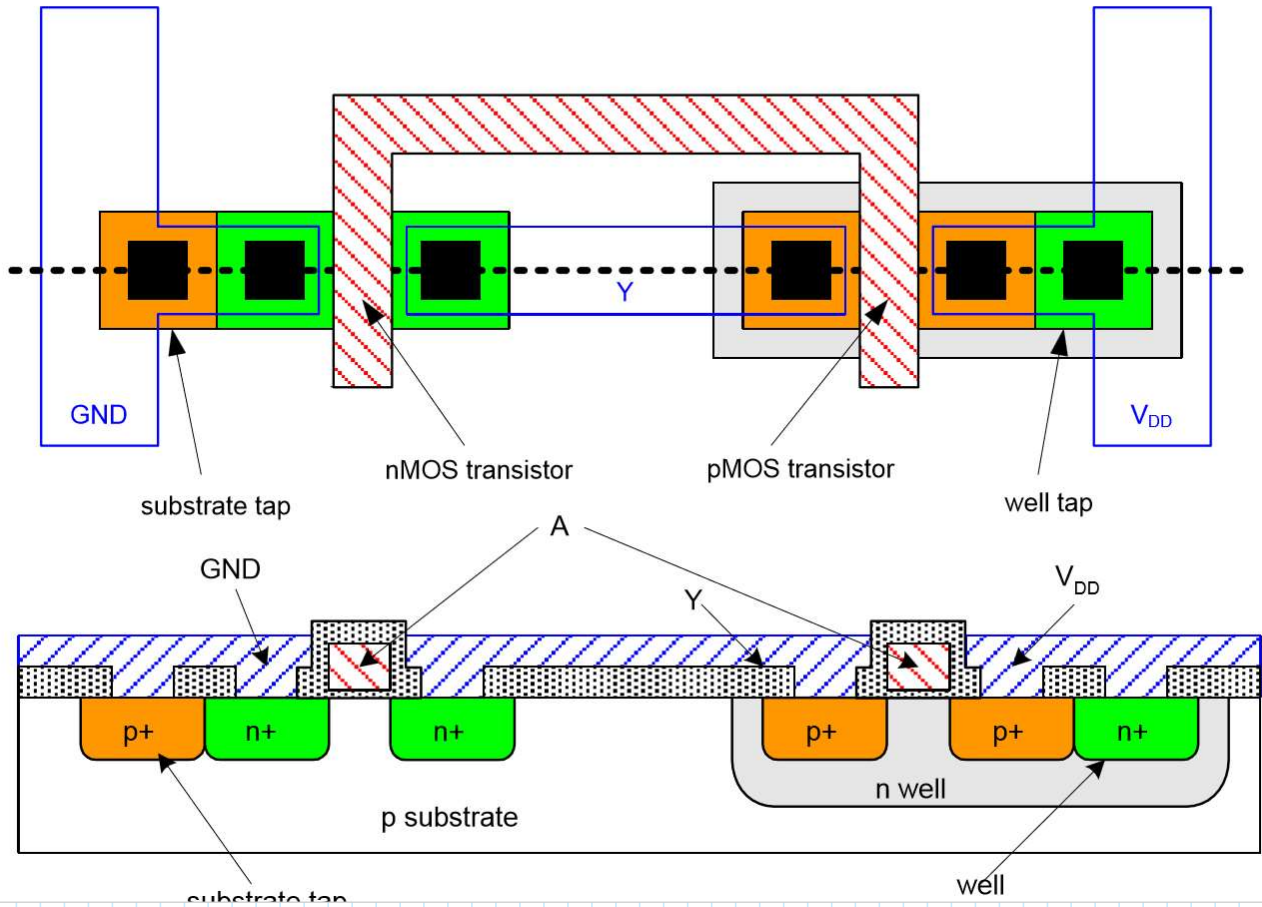


Inverter Layout

- ❑ Transistors and wires are defined by *masks*
- ❑ Cross-section taken along dashed line



Layout vs Cross-Section

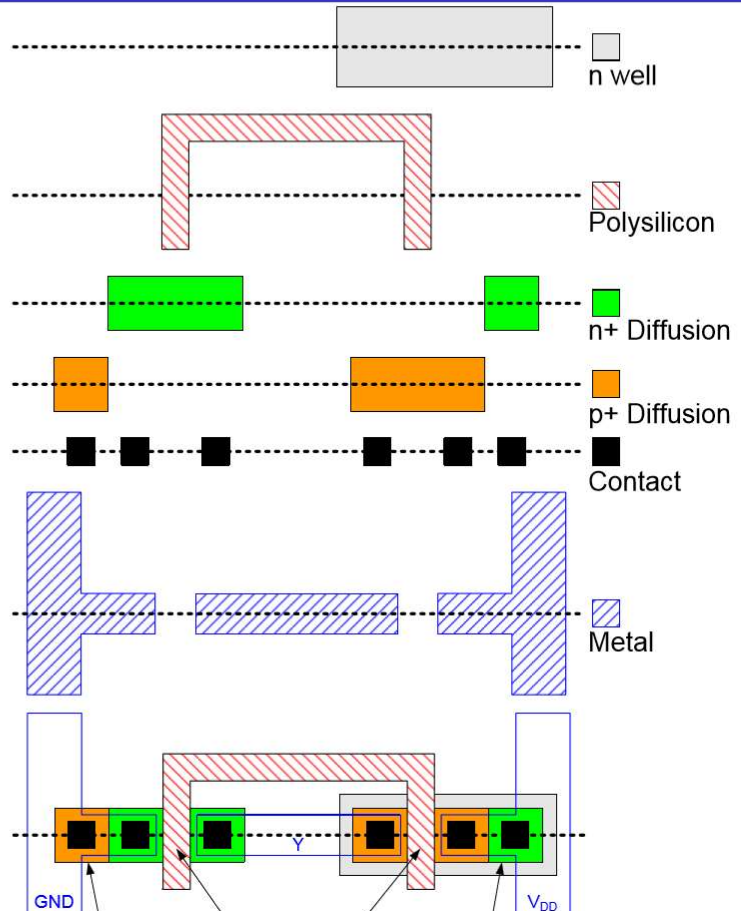


Inverter Mask Set

❑ Six masks

1. N-well
2. Polysilicon
3. n+ diffusion
4. p+ diffusion
5. Contact
6. Metal

❑ **The transistor is formed by the intersection of polysilicon and diffusion**



Fabrication

- ❑ Chips are built in huge factories called fabs (fabrication plants)
- ❑ Fabs contain “clean rooms” as large as football fields
- ❑ A fab processing 300 mm wafers in a 45 nm process costs about \$3 billion (that’s why many semiconductor companies are fabless)
- ❑ Big players: Intel, Global Foundries, TSMC, UMC, IBM, ...



Robustness

- ❑ A central challenge in building integrated circuits is to get millions or billions of transistors to all function
 - Not just once, but for a quintillion consecutive cycles
- ❑ Despite challenges, engineers routinely build robust integrated circuits
 - lifetimes exceed ten years of continuous operation
- ❑ Conventional static CMOS circuits are exceptionally well-suited to the task
 - have great noise margins
 - are minimally sensitive to variations in transistor parameters
 - will eventually recover even if a noise event occurs

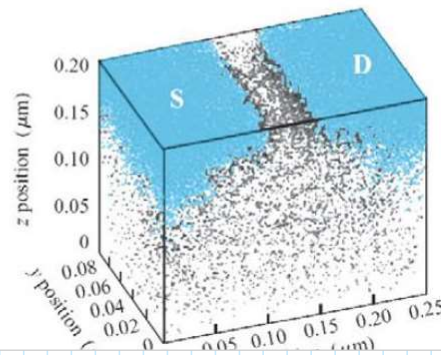
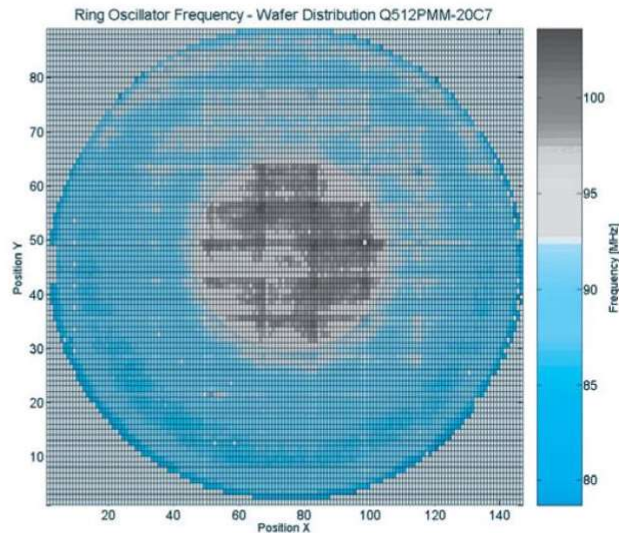
Process Variations

❑ Examples of process variations

- Channel length
 - Lithography limitations
 - Varying etch rates
 - Line edge roughness
- Threshold voltage
 - Random dopant fluctuations
 - High V_t has higher doping and higher variations

❑ E.g., intel CPU binning

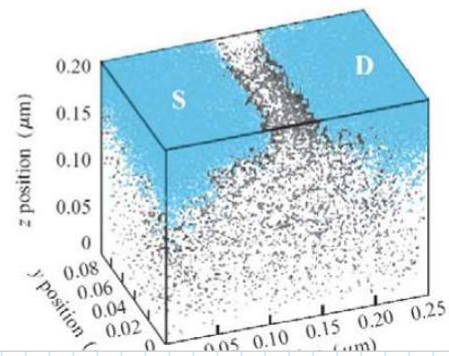
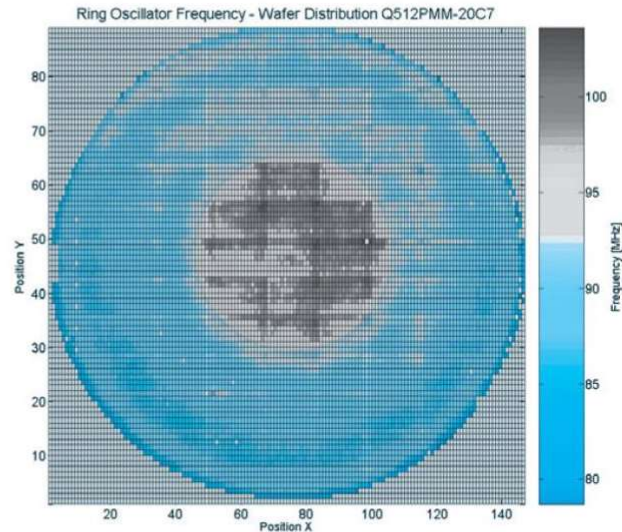
- 1.8 GHz, 2.0 GHz, 2.2 GHz
- i3, i5, i7



Process Variations

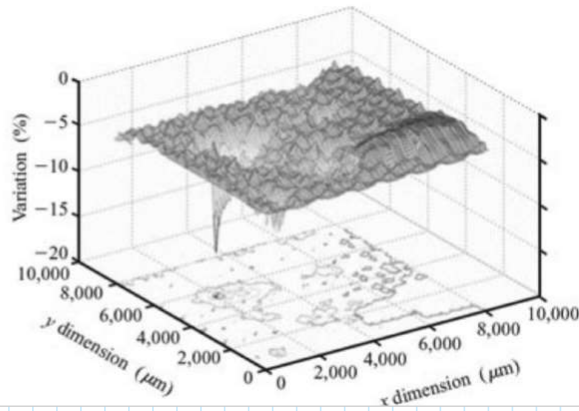
❑ Process Variations Classification

- Lot-to-lot (L2L)
- Wafer-to-wafer (W2W)
- Die-to-die (D2D) or inter-die
- Within-die (WID) or intra-die
 - Most important for analog
 - a.k.a. mismatch



Voltage Variations

- ❑ Supply voltage may vary around its nominal value
 - Regulator tolerances
 - IR drops
 - di/dt noise
- ❑ Voltage varies in both space (across chip) and time
- ❑ Typically tolerate 10% variation
 - To first order, leads to 10% delay variations
- ❑ Voltage droop map:



Temperature Variations

- ❑ Drain current decreases with temperature
- ❑ Ambient temp ranges:
 - Commercial (0 to 70°C)
 - Industrial (-40 to 85°C)
 - Military (-55 to 125°C)
- ❑ Junction temp may significantly exceed the ambient
 - Function of power consumption and package thermal resistance
 - **Commercial parts commonly verified at 125°C junction temp**
- ❑ Temperature varies in both space (across chip, a.k.a. temperature gradients) and time (temperature fluctuations)
 - Circuits in a 1 mm diameter see nearly the same temperature
 - Temperature varies in time on a scale of milliseconds

Design Corners

- ❑ MOS: Slow (S), typical or nominal (T), and fast (F)
- ❑ Voltage: S ($0.9V_{DD}$), T (V_{DD}), and F ($1.1V_{DD}$)
- ❑ Temp: S (125°C), T (70°C), and F (0°C)
- ❑ Binning: Faster parts are rated for higher frequency and sold for more money

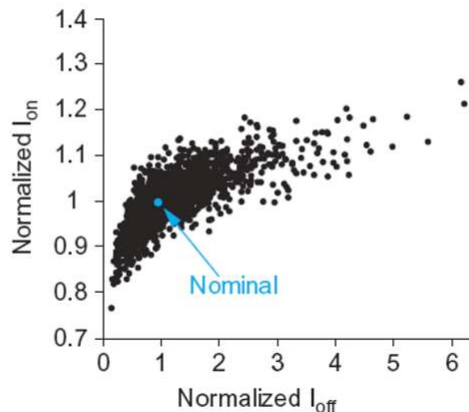


Design Corners

- ❑ Few corners for old technologies: simulate all
- ❑ Thousands of corners for DSM nodes: sophisticated CAD tools are required to identify what corners are important for a given circuit
- ❑ Burn-in corner during testing
 - 125 to 140 °C externally, corresponding to an even higher internal temperature
 - 1.3–1.7× nominal VDD
 - Very high leakage

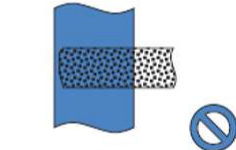
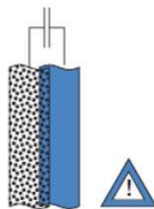
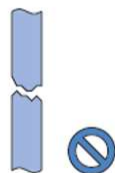
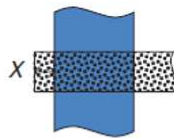
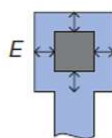
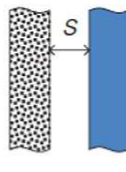
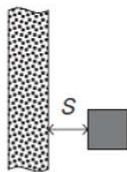
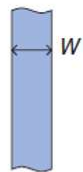
Monte Carlo Simulation

- ❑ As process variation increases, the worst-case corners become too pessimistic for practical design
- ❑ Monte Carlo: repeated simulations with parameters randomly varied each time
- ❑ Look at scatter plot of results to predict yield
- ❑ Ex: impact of V_t variation
 - ON-current
 - leakage



Layout Design Rules (DRs)

- ❑ The contract between the designer and the process engineer
- ❑ Same layer: min width and min spacing
- ❑ Different layers: min spacing, min enclosure (overlap), and min extension



a) min. width

b) min.intralayer spacing

c) min. interlayer spacing

d) min. enclosure

e) min. extension

Layout DRs cont'd

- ❑ Real-life layout DRs are very complicated
 - Checked by a software CAD tool
 - DRC: Design rule check
 - Mentor Calibre is the industry standard DRC tool
 - Checks the layout using a “rule file” (a.k.a. rule deck) and generates a list of DRC errors

Layout DRs cont'd

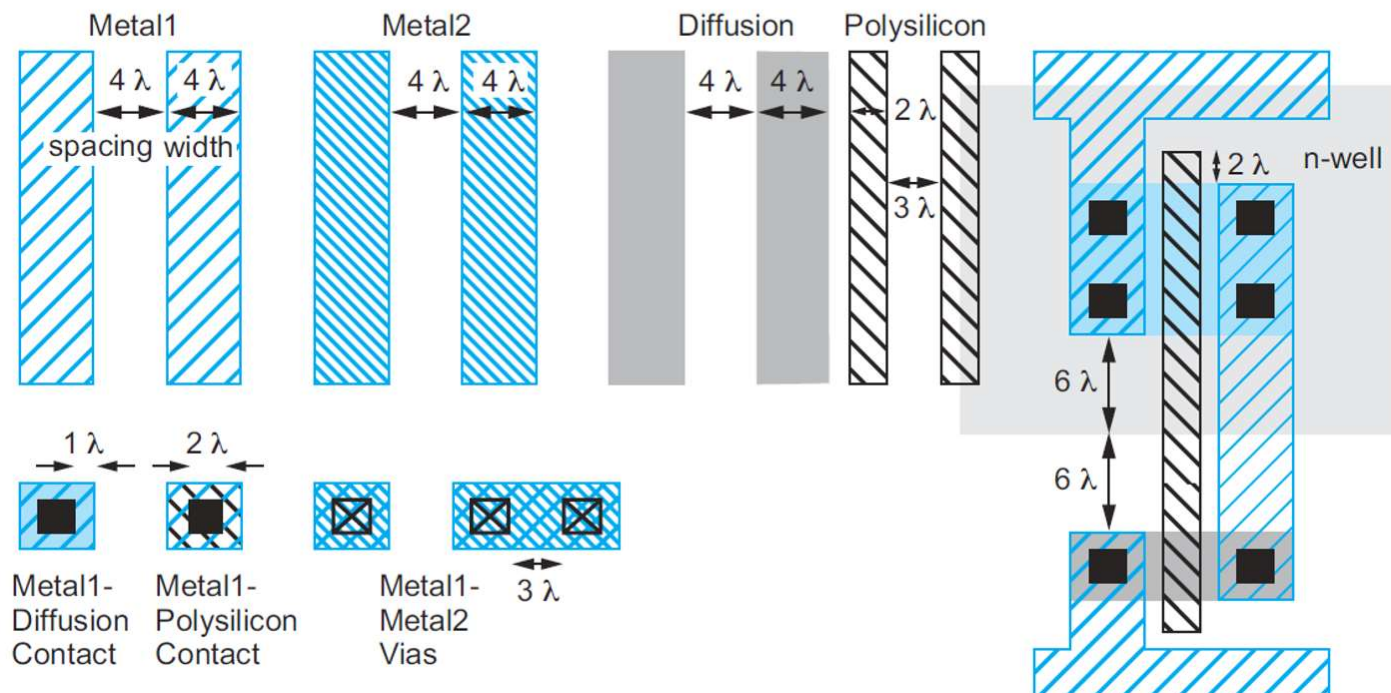
- ❑ Generally, the foundry will not accept any design with DRC errors
 - All errors must be fixed
 - Sometimes the errors are not real and can be ignored (you must check with the foundry)
 - In special cases, minor DRC errors can be accepted by the foundry (but you must request an official DRC error waiver)

λ -Based Layout DRs

- ❑ Simplified scalable conservative rules to get you started
- ❑ Feature size $f = 2\lambda$ = distance between source and drain
 - Set by minimum width of polysilicon
- ❑ Feature size improves 30% every 3 years or so
- ❑ Normalize layout design rules in terms of $\lambda = f/2$
 - E.g. $\lambda = 0.3 \mu\text{m}$ in $0.6 \mu\text{m}$ process

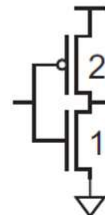
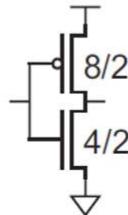
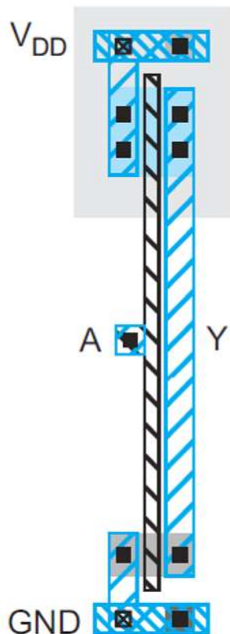
λ -Based Layout DRs cont'd

- ❑ Simplified scalable conservative rules to get you started
- ❑ Real-life DRs are in microns and are much more complex



Inverter Layout

- ❑ Transistor dimensions specified as Width / Length
 - Minimum size is $4\lambda / 2\lambda$, sometimes called 1 unit
 - In $f = 0.6\ \mu\text{m}$ process, this is $1.2\ \mu\text{m}$ wide, $0.6\ \mu\text{m}$ long

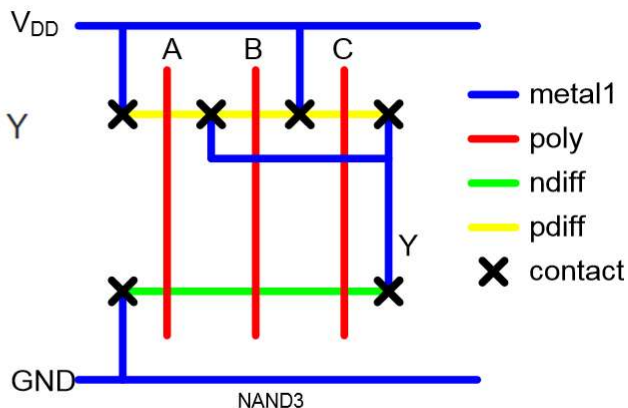
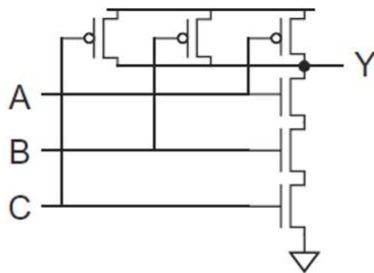
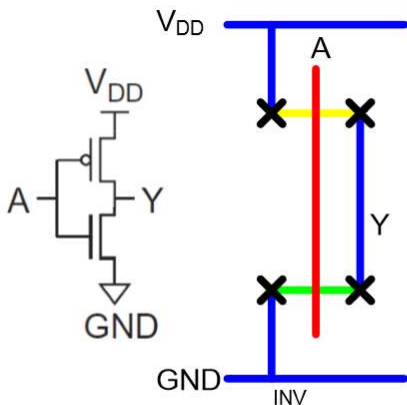


Gate Layout

- ❑ Layout can be very time consuming
 - Design gates to fit together nicely
 - Build a library of standard cells
- ❑ Line of diffusion style
 - Simple and straightforward layout style for standard cells
 - **Four horizontal strips:**
 1. metal ground at the bottom
 2. n-diffusion
 3. p-diffusion
 4. metal power at the top
 - Polysilicon lines run vertically to form transistor gates
 - Metal wires connect the transistors appropriately

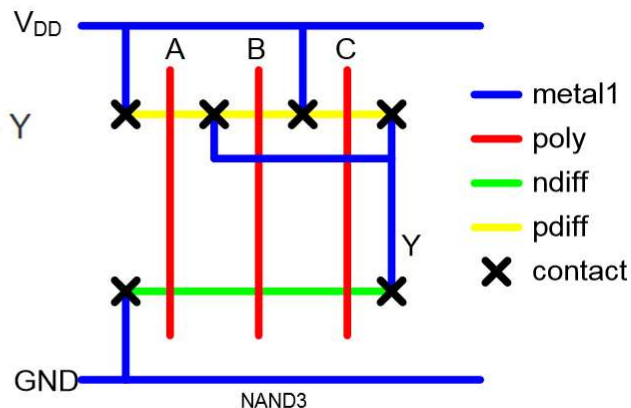
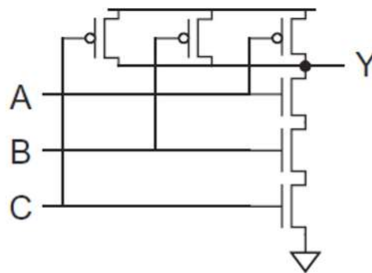
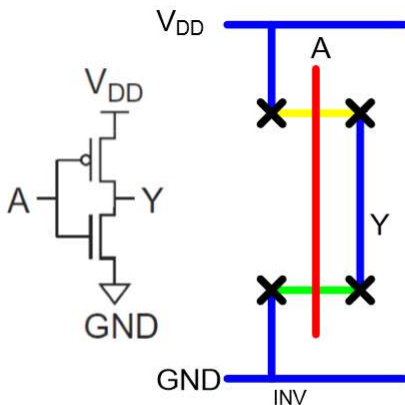
Stick Diagrams

- ❑ *Stick diagrams* help plan layout quickly
 - Need not be to scale
 - Draw with color pencils or dry-erase markers
- ❑ Transistor = intersection of poly and diffusion
- ❑ **Note source and drain sharing**



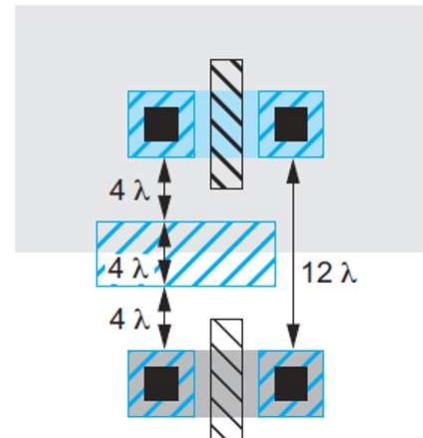
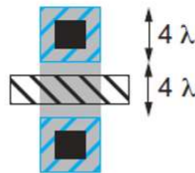
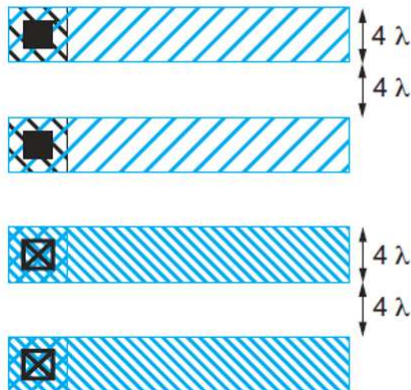
Stick Diagrams cont'd

- ❑ Metal1 GND rail at bottom
- ❑ Horizontal N-diffusion and p-diffusion strips
- ❑ Metal1 V_{DD} rail at top
- ❑ Vertical polysilicon gates
- ❑ Metal1 makes connections



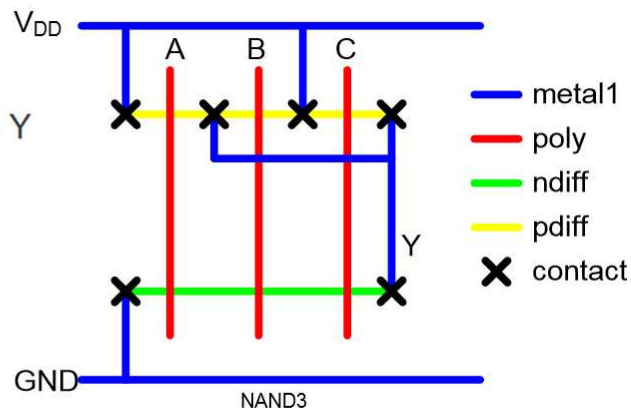
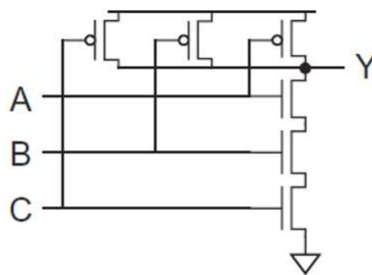
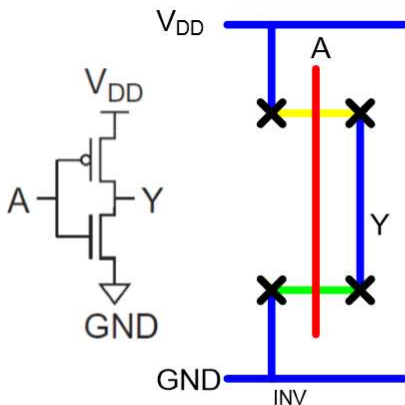
Area Estimation (Using λ -Based DRs)

- ❑ A *wiring track (WT)* is the space required for a wire
 - 4λ width, 4λ spacing from neighbor = 8λ pitch
- ❑ Transistors also consume one WT (if W is min)
- ❑ Wells must surround transistors by 6λ
 - Implies 12λ between opposite transistor flavors
 - Leaves room for one WT $\rightarrow$ total equivalent to 3 WT (if W is min)



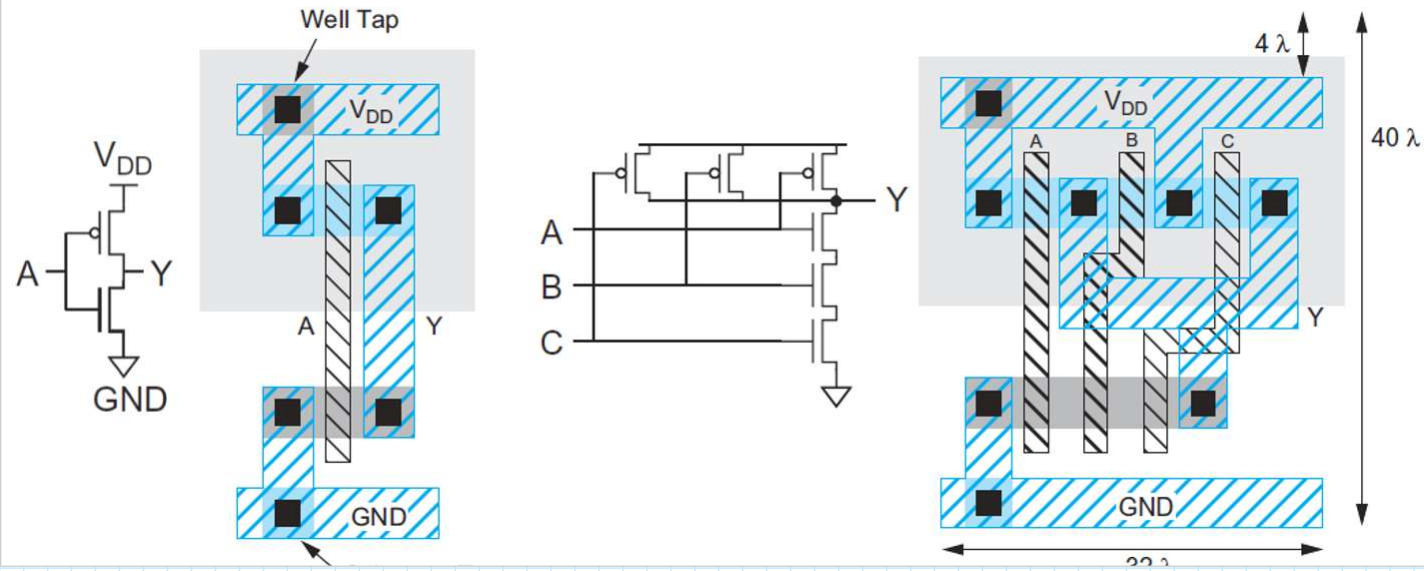
Area Estimation cont'd

- ❑ Estimate area by counting wiring tracks (WT)
 - Multiply by 8 to express in λ
- ❑ For the inverter: 2 WT by 5 WT $\rightarrow 16 \lambda$ by 40λ
- ❑ For the NAND3: 4 WT by 5 WT $\rightarrow 32 \lambda$ by 40λ



INV and NAND3 Cells

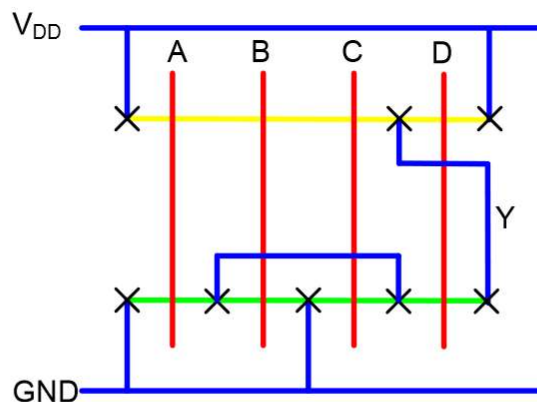
- ❑ INV area: 16λ by $40\lambda \rightarrow$ exactly as estimated!
- ❑ NAND3 area: 32λ by $40\lambda \rightarrow$ exactly as estimated!
- ❑ **Note source and drain sharing**
- ❑ Practically: Use **MANY** substrate and well taps



Example: O3AI

- Estimate the cell width and height of O3AI cell using min size transistors

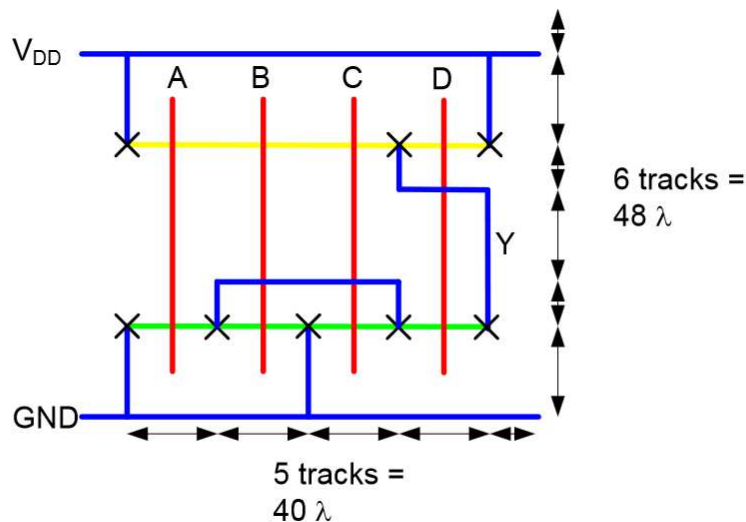
$$Y = \overline{(A + B + C) \cdot D}$$



Example: O3AI

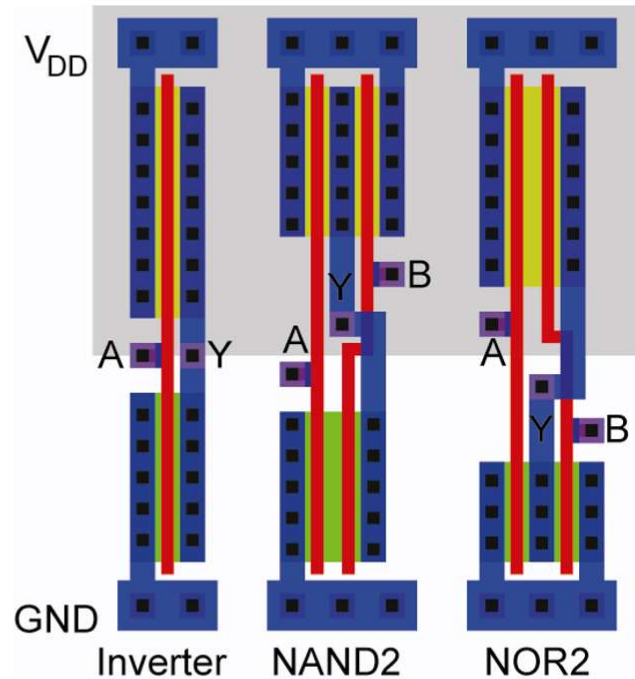
- Estimate the cell width and height of O3AI cell using min size transistors

$$Y = \overline{(A + B + C) \cdot D}$$

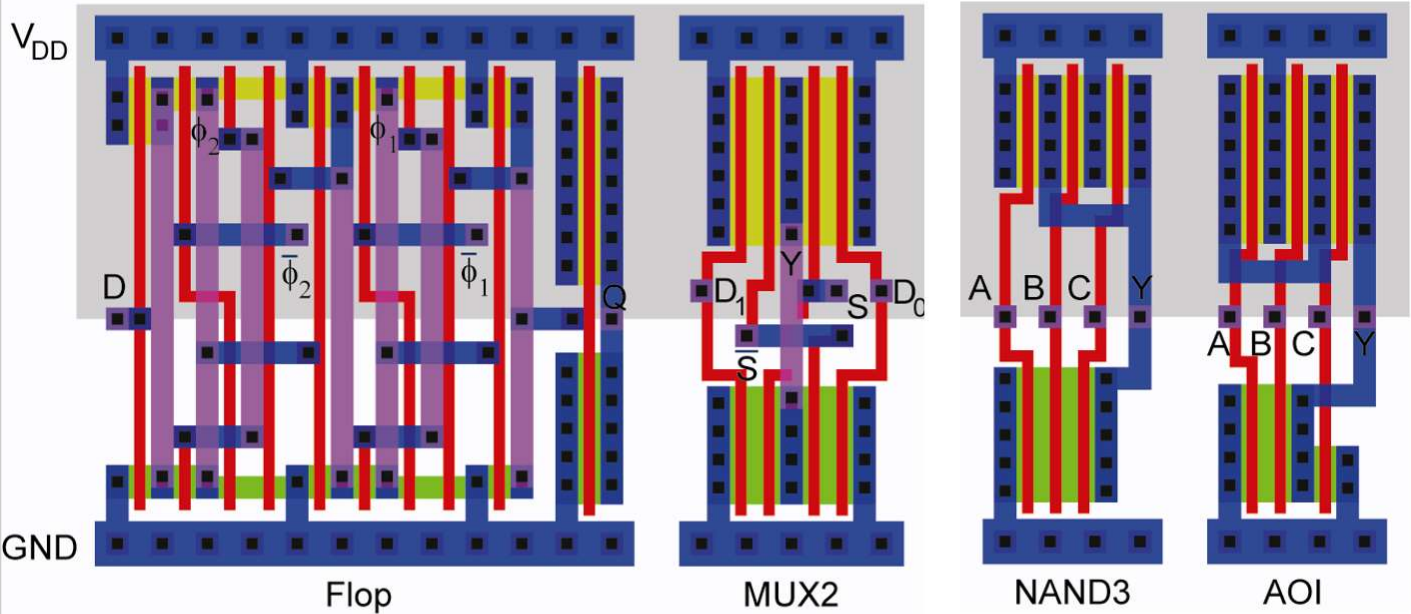


Standard Cell Design

- Standard cell design methodology
 - V_{DD} and GND supply rails in M1 (uniform cell height)
 - nMOS at bottom and pMOS at top (uniform well height)
 - Adjacent gates should satisfy design rules
 - All gates include well and substrate contacts
 - Cells connected by abutment

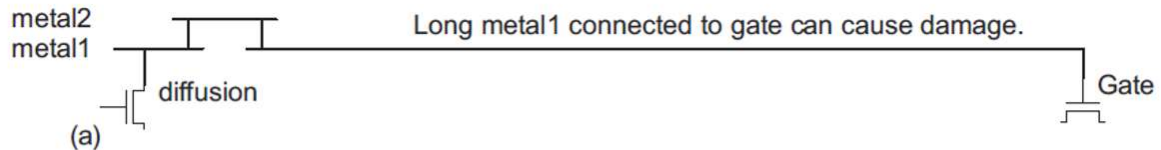


Standard Cell Design cont'd



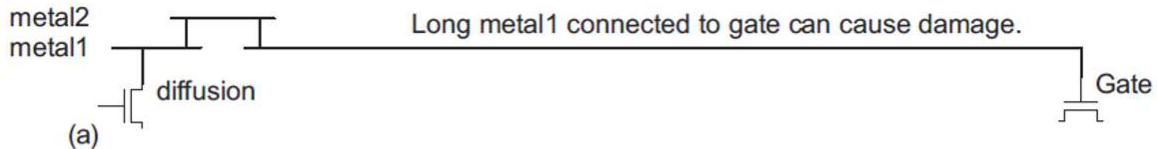
Antenna Rules

- ❑ Antenna effect:
 - Metal wires charge up when it is plasma-etched
 - May damage thin gate oxide (plasma-induced gate-oxide damage)
- ❑ Antenna rules define max metal area to gate oxide area to avoid damage



Antenna Rules cont'd

- Ways to fix it:
- Jump to upper metal layer just before the gate
 - Add a reverse-biased diode (bleeds off the charges during the high-temperature plasma-etch process)
 - D/S diffusion is already a reverse-biased diode (if M1 is used)

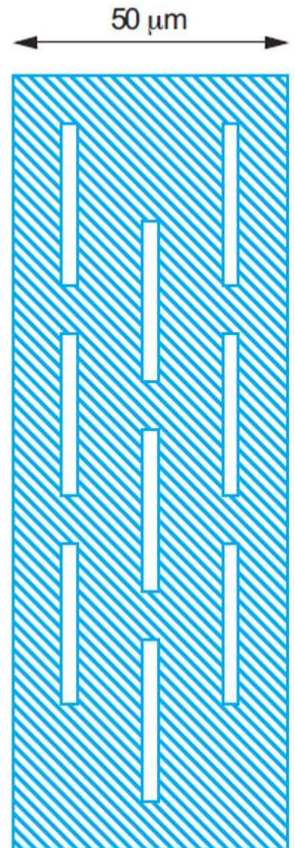


Layer Density Rules

- ❑ Each layer must have adequate density to ensure uniform etching and proper CMP
- ❑ A metal layer may have to satisfy minimum density (e.g., 30%) and maximum density (e.g., 70%)
- ❑ There are both global (whole chip) and local (e.g., $100\mu\text{m} \times 100\mu\text{m}$ area) density rules
- ❑ If not already satisfied by the design dummy structures (and holes) must be added to fix the errors
- ❑ Foundries usually provide a “metal-fill” script to add dummy structures (and holes) automatically
 - May add parasitic capacitances or cause mismatch
 - Should be disabled over sensitive analog circuitry (by using “fill-stop” layout layer)

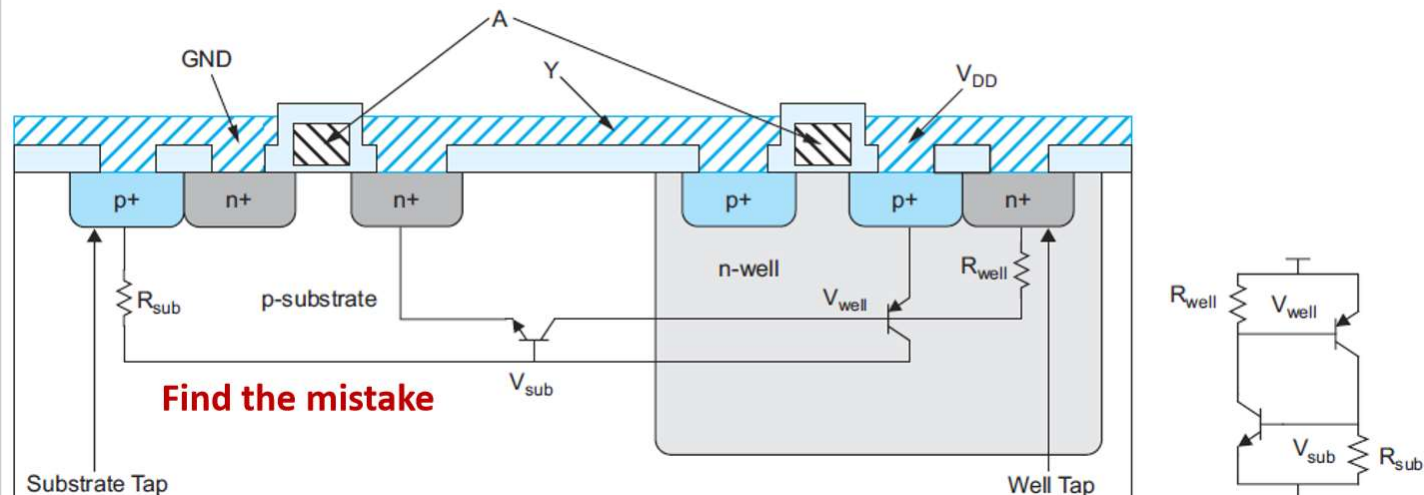
Metal Slotting Rules

- ❑ Wide metal wires are usually used for global wires (e.g., power routing) to decrease resistance
- ❑ Slots must be added parallel to the current flow to provide stress relief and ensure proper fabrication
- ❑ Foundries usually provide a script to automate this process



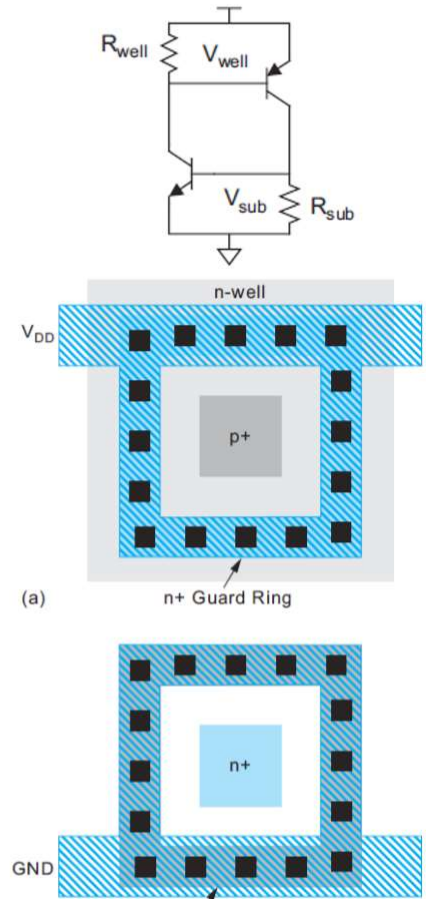
Latchup

- ❑ Ordinarily, both parasitic bipolar transistors are OFF.
- ❑ If substantial current flows in the substrate:
 - $V_{\text{sub}} \uparrow$ turning ON the npn transistor
 - $V_{\text{well}} \downarrow$ turning ON the pnp-transistor
- ❑ A positive feedback loop is triggered with a large current flowing between VDD and GND



Latchup

- ❑ Latchup prevention is easily accomplished by minimizing R_{sub} and R_{well}
 - Use MANY substrate/well taps close to each transistor
- ❑ I/O pads are more prone to latchup because external voltages can ring below GND or above VDD
 - Guard rings should be used (a low resistance path to collect stray currents)
- ❑ Latchup is not important for:
 - SOI processes (no parasitic BJT)
 - DSM nodes with $V_{DD} < 1.4$ (no enough voltage to turn on two BJTs)

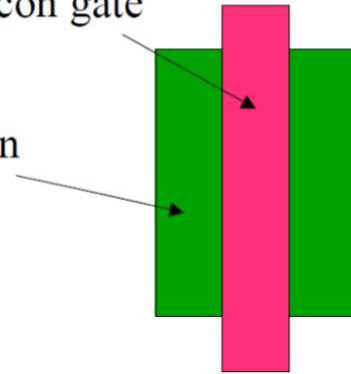


Transistor Layout

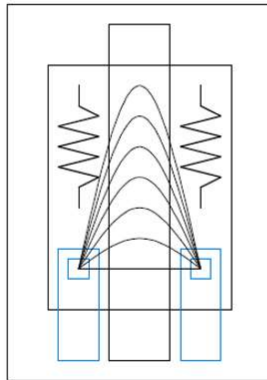
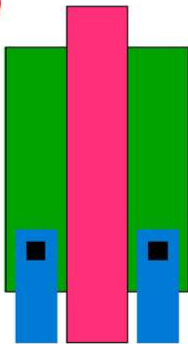
- ☐ The transistor is formed by the intersection of polysilicon and diffusion
- ☐ Add GOOD contacts to drain and source and source
- ☐ Add GOOD and NEARBY substrate/well contacts.

Polysilicon gate

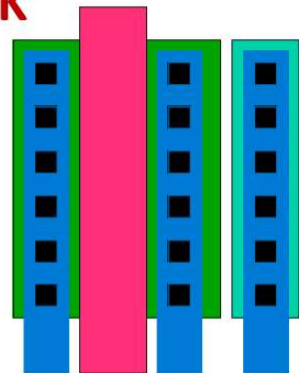
diffusion



BAD

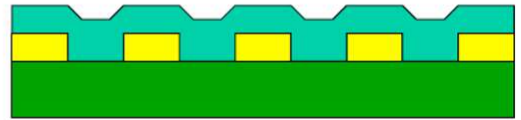


OK



Multiple or Single Contact/Via?

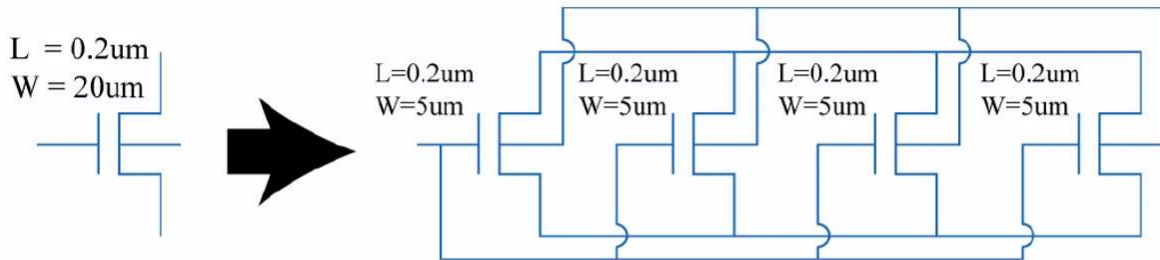
- ❑ Curvature in the metal layer can cause fractures



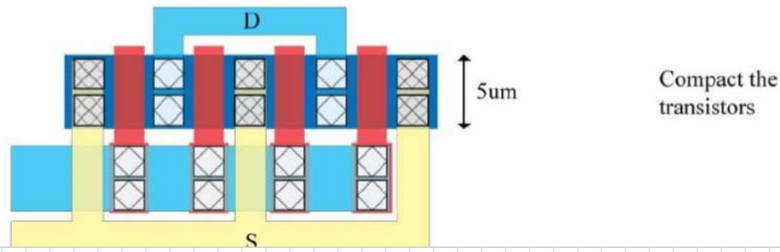
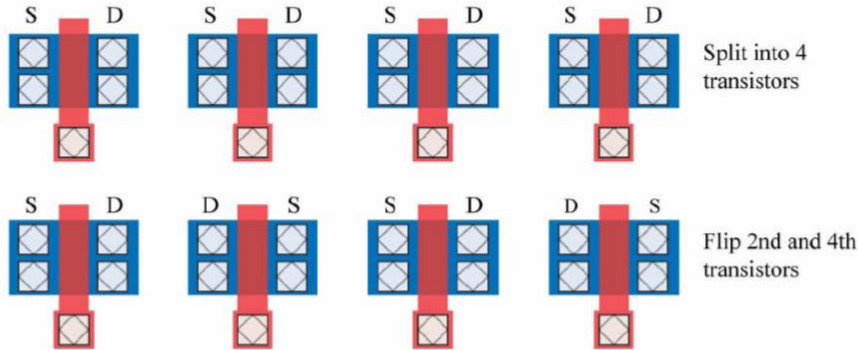
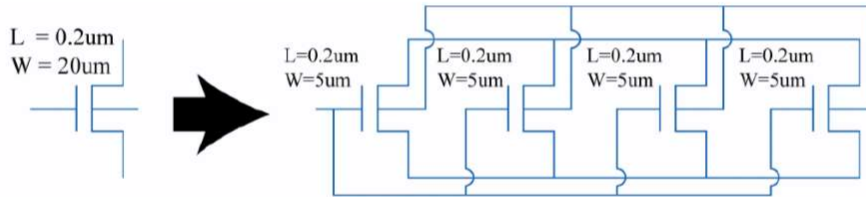
Reliability problems, possible electro-migration

Multi-finger Transistor Layout

- ❑ Analog transistors often have large W/L ratio
- ❑ Multi-finger layout is commonly used
 - Compact layout
 - Smaller parasitic capacitances
 - Smaller gate resistance

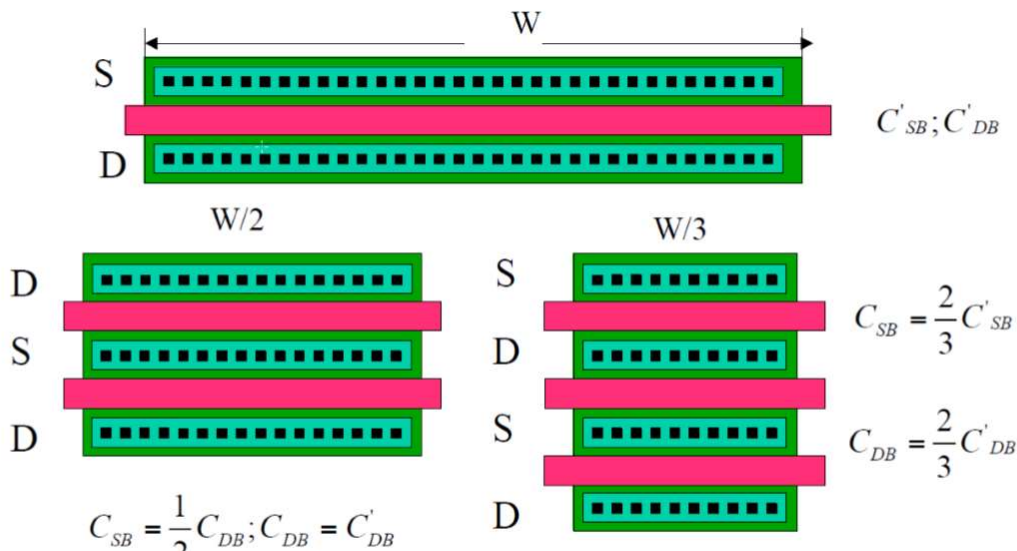


Multi-finger Transistor Layout



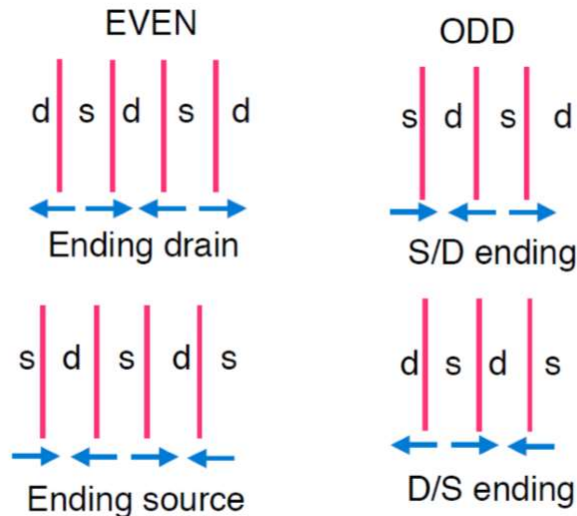
Multi-finger Transistor Layout

- ❑ Analog transistors often have large W/L ratio
- ❑ Multi-finger layout is commonly used
 - Compact layout
 - Smaller parasitic capacitances
 - Smaller gate resistance

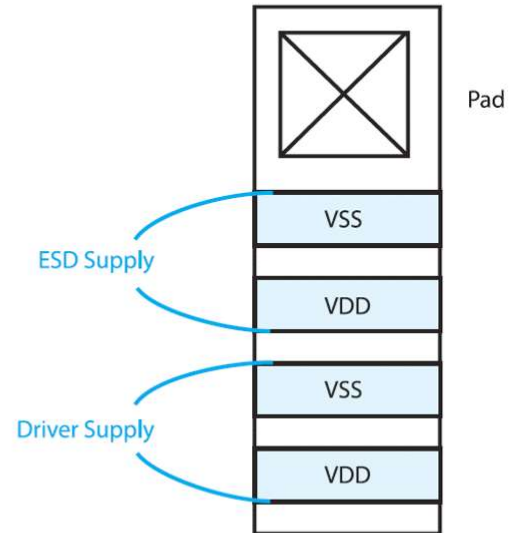
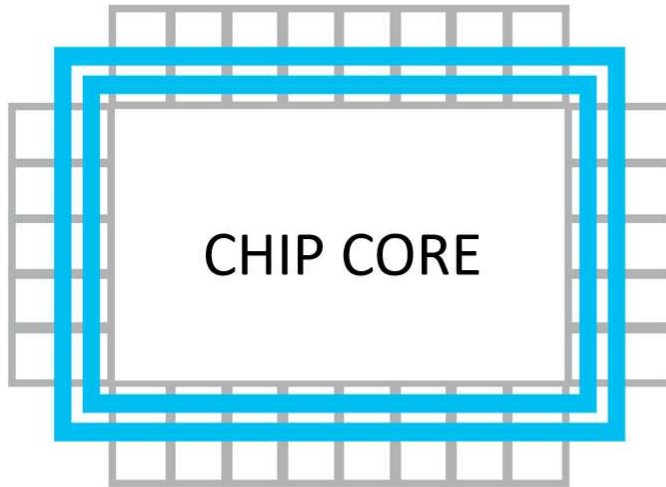


Multi-finger Options

- ❑ Usually use even no. of fingers with source ending
 - Decrease drain capacitance
 - Also easier if the source is to be connected to bulk (gnd)
- ❑ But may need different combinations when stacking different transistors (more on this later)

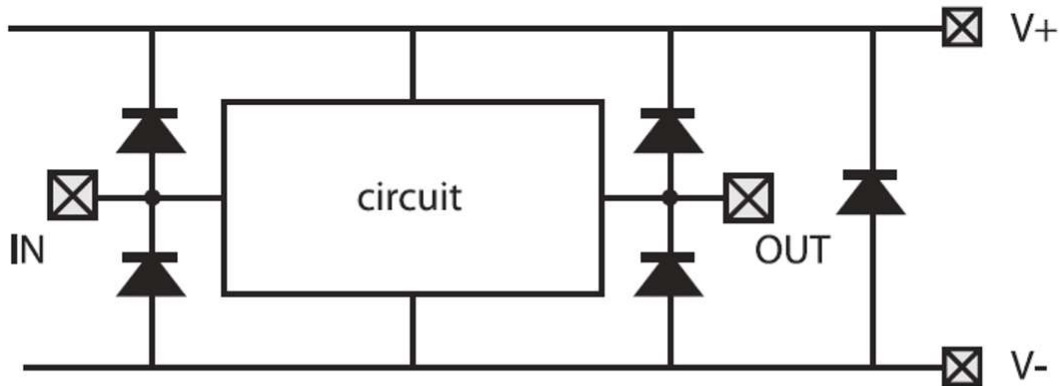


Padframe and IO Cells



ESD Protection

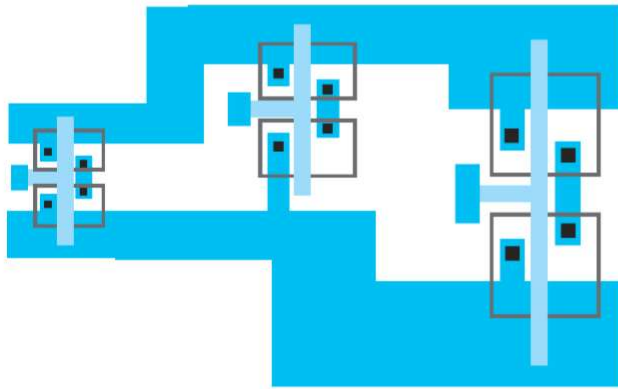
- ❑ Any ESD current will flow in the ESD diodes instead of flowing into the circuit
- ❑ The diodes should be large enough to handle high current spikes



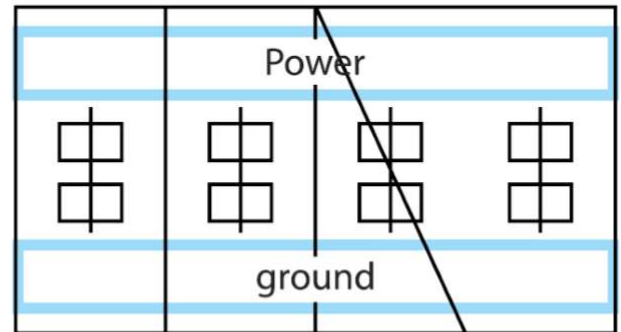
Digital Layout

- ❑ Fixed height, variable width
- ❑ Continuous VDD, GND, and NWELL

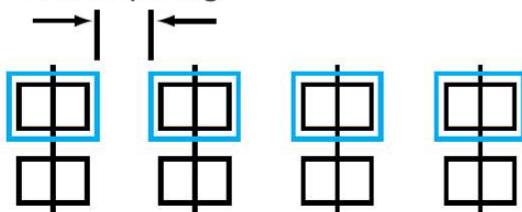
Bad



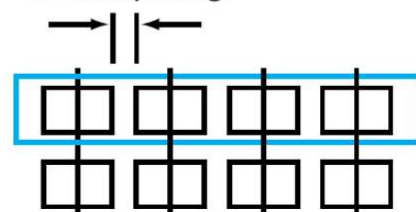
Good



N well spacing

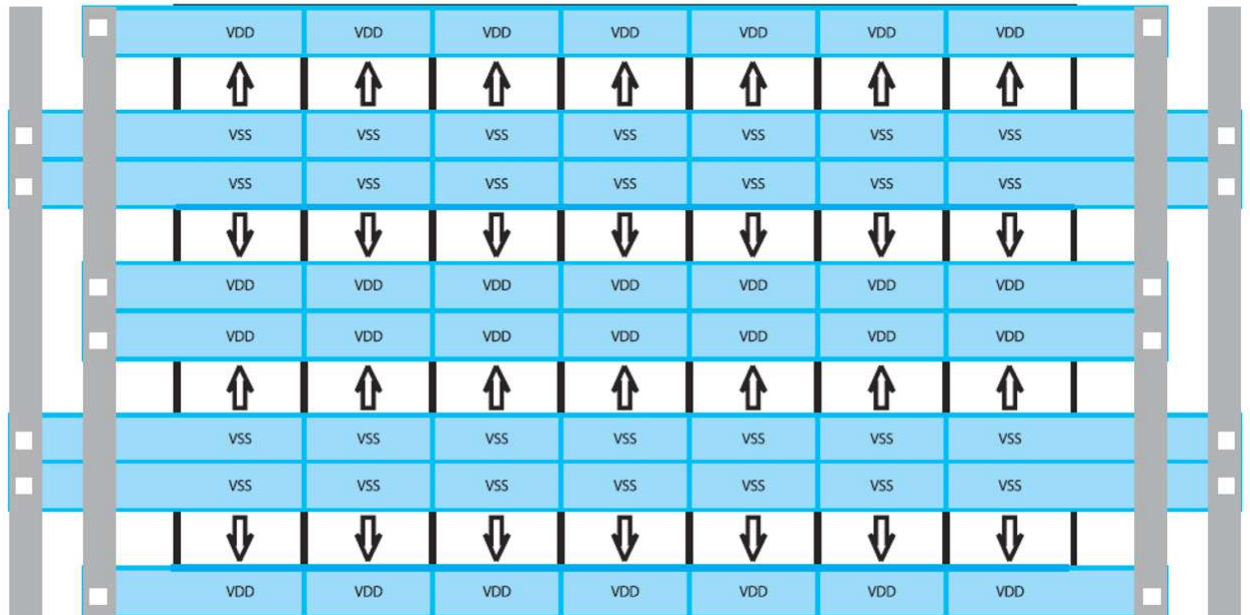


device spacing



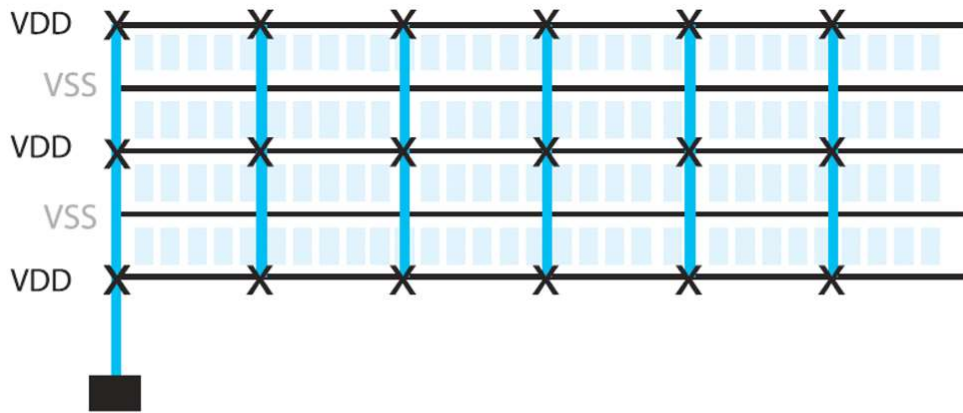
Digital Layout

- ❑ Fixed height, variable width
- ❑ Share power rails between flipped rows
- ❑ No need for routing channels in modern technologies



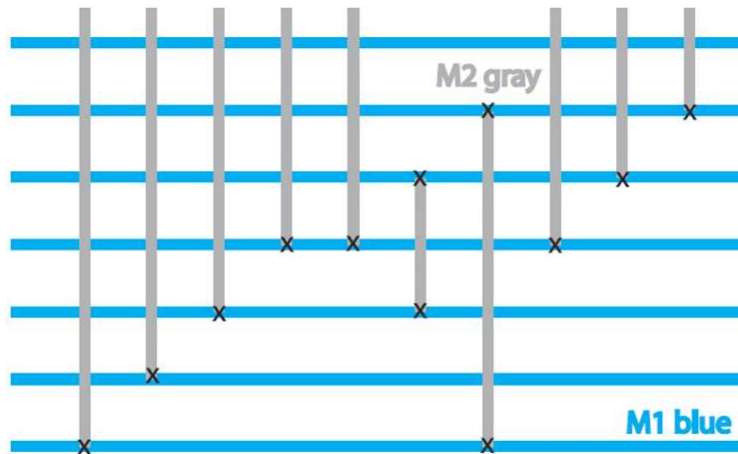
Strapping Power Rails

- ❑ Distant cells see more resistance through the rails due to the length of wire
- ❑ By laying straps of metal across your power rails the overall resistance reduces



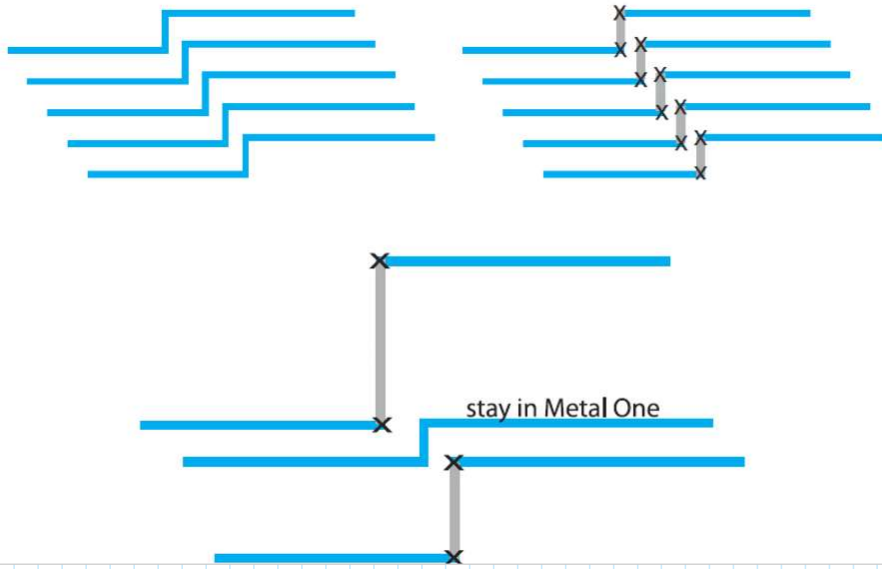
Directional Layer Technique

- ❑ Metal One runs horizontally and Metal Two runs vertically
- ❑ Allows wires to run in complex knots using few metal layers without becoming trapped
- ❑ Common for digital signals, but is not strictly followed for sensitive analog/RF signals



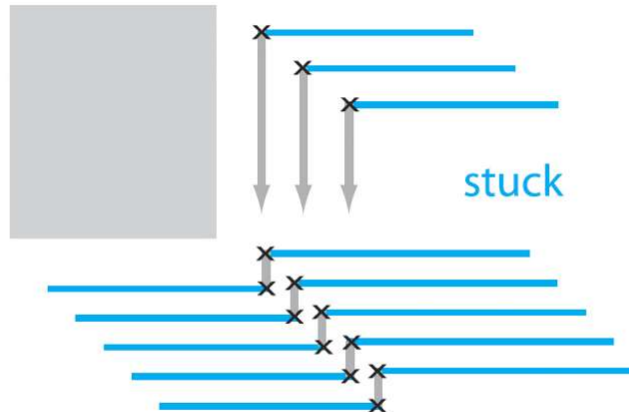
Directional Layer Technique

- ❑ This rule is not useful for short jumps
 - Via resistance
 - Blocking other wires
- ❑ Do NOT change metal layer for short jumps



Directional Layer Technique

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 - Via resistance
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Digital vs Analog Layout

❑ Digital layout

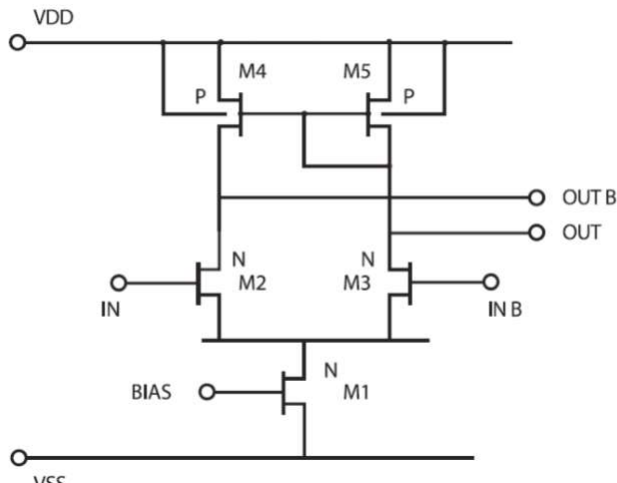
- Primary objective is optimizing density
- Millions of gates
- Strict standard cell rules

❑ Analog layout

- Primary objective is matching and performance
- Few amplifiers
- Many options instead of many rules
- Strong interaction required between circuit and layout designers

Layout Designer Questions

- ☐ What does this circuit do (function, frequency, precision, etc.)?
- ☐ How much current does it take (current density, min wire width)?
- ☐ Where are the high and low current paths (not all wires will carry high current)?
- ☐ What matching requirements are there?
- ☐ Is there anything else?



Layout Designer Questions

What Does It Do?

Circuit Function?

What Frequency?

Low Parasitic Nets?

How Much Current?

Where Is High Current Flow?

Where Is Low Current Flow?

Does Current Flow to Other Blocks?

What Matching is Required?

Identify Nets

Identify Transistors

Identify Other Blocks

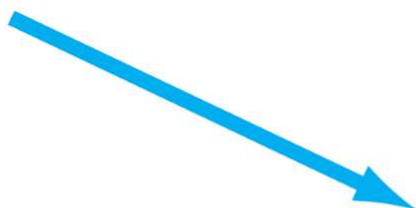
Identify Any Remote Components

Anything Else?

Device Placement Concerns?

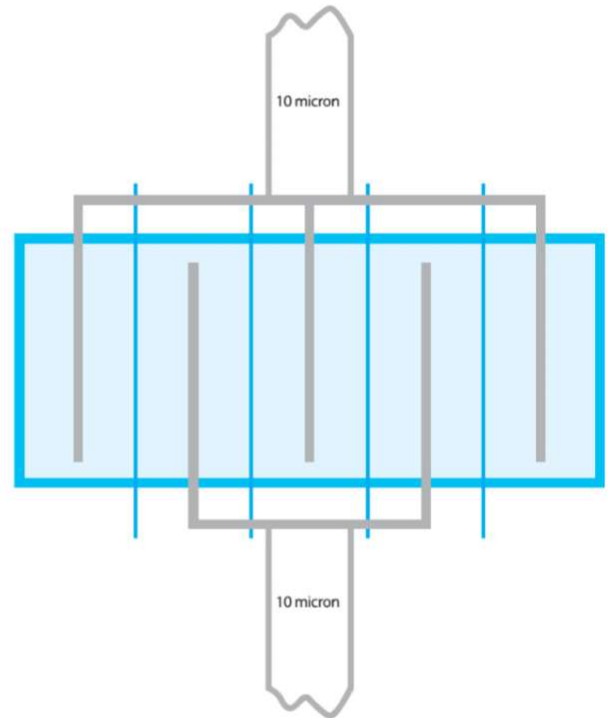
Metal Choices?

Isolation Requirements?



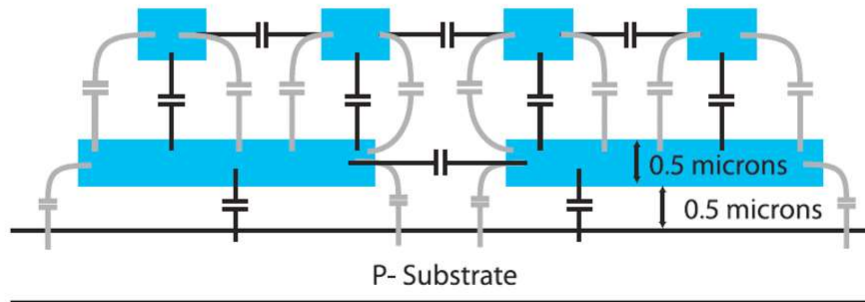
Ex: Routing with Multi-finger Device

- ❑ Route from center not from sides
 - Current spread out evenly
- ❑ Ex: four-finger MOSFET
 - 4 mA current
 - 0.5 mA/ μm max density
 - 8 μm min wire
 - 3 source regions
 - 2 drain regions
 - Source/drain metallization must handle the required current



Parasitic Capacitance

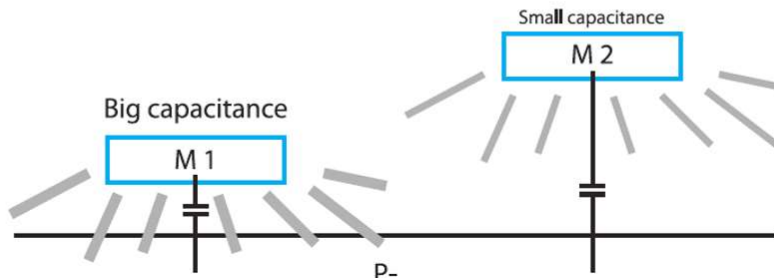
- ❑ Capacitance in the order of fF
- ❑ Important for high frequency and precision circuits
 - Use short wire length
 - Use isolated wire (far from everything else)
- ❑ You have to look at metal height from subs, width, and thickness



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Metal	M1	M2	M3	M4
Min. Width	0.8	0.8	2.4	6.5
Cap/Unit Area (fF/ μm^2)	5	3	2.5	1.5



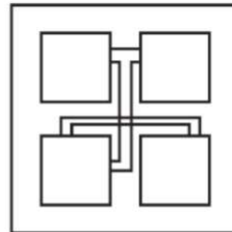
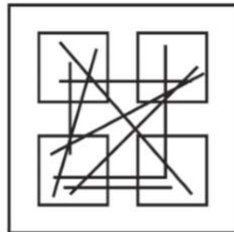
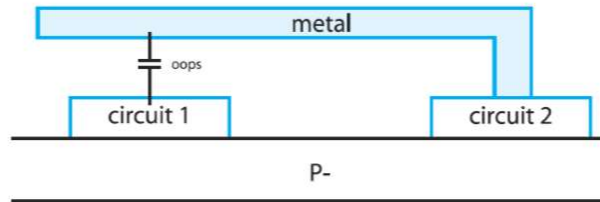
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Min. Width	0.8	0.8	2.4	6.5
Cap/Unit Area (fF/ μm^2)	5	3	2.5	1.5
Capacitance 10- μm wire	40	24	66	97.5

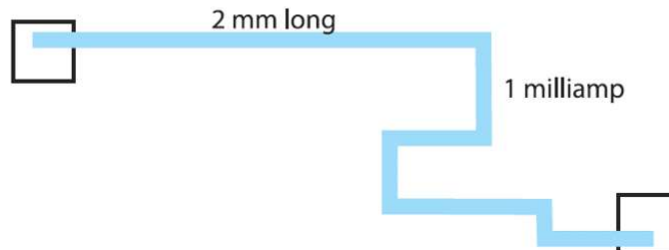
Parasitic Capacitance

- ❑ Avoid Routing over Circuitry
 - Not important in digital, but may affect sensitive analog blocks/signals
- ❑ A good floorplan will help



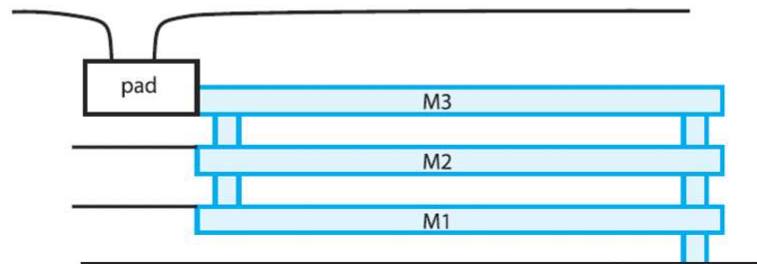
Parasitic Resistance

- ❑ May cause large $I \cdot R$ drops
- ❑ Ex:
 - 2mm long / 2 μm width = 1000 squares
 - $R = 1000 \text{ squares} \cdot 50 \text{ m}\Omega/\text{square} = 50 \text{ }\Omega$
 - $V_{\text{drop}} = I \cdot R = 50 \text{ mV} \rightarrow$ definitely not acceptable
- ❑ Increase wire width to reduce resistance
 - Max current density is not the only thing to worry about
- ❑ Or use thick metal in higher layers
- ❑ Or stack wires to save space



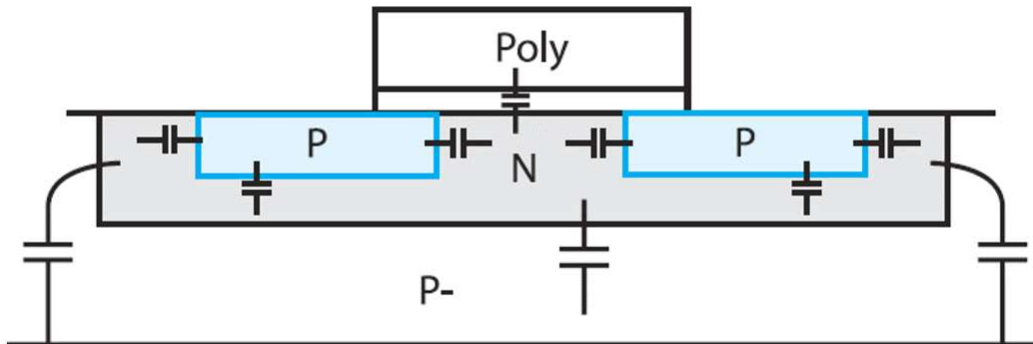
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MOS Capacitance/Resistance

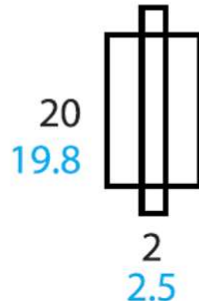
- ❑ Use multi-finger layout
 - Reduces gate resistance (multiple gate stripes in parallel)
 - Splitting the device into two reduces RC time constant by 4
 - Drain/source sharing reduces capacitance



Matching: Orientation

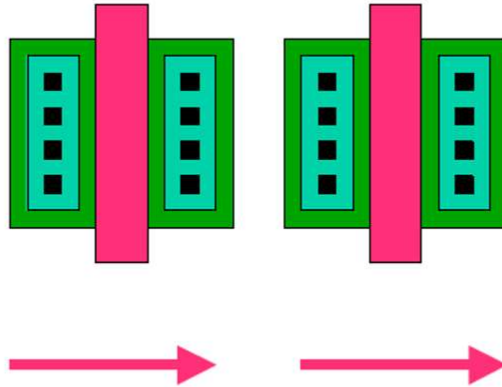
- ❑ orientation-specific processing errors will severely degrade matching
- ❑ Use the same orientation for matched devices
- ❑ Also good practice to use the same orientation all over the chip
 - Now a design rule for FinFET

20 x 2	match as drawn	20 x 2
19.8 x 2.5	no match after processing	20.5 x 1.8



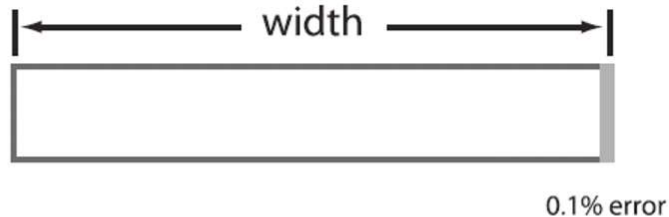
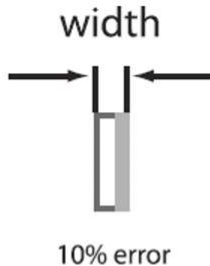
Matching: Orientation

- ❑ Current should be flowing in same direction



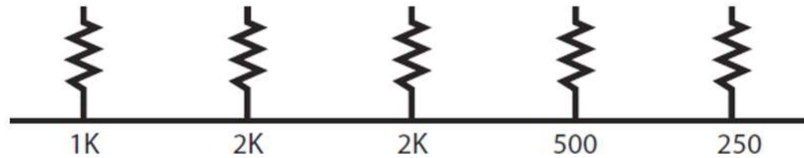
Matching: Unit Elements

- ❑ Applies to all components: transistors, resistors, capacitors, etc.

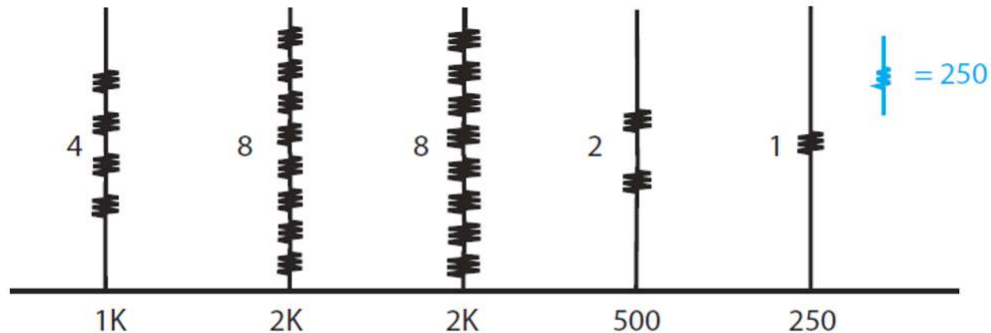


Matching: Unit Elements

☐ Terrible

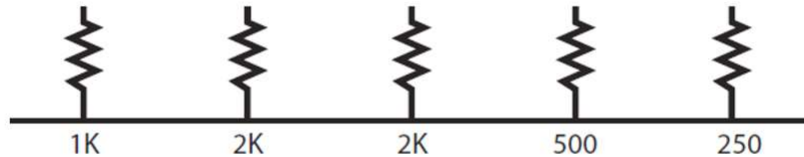


☐ Good (use lowest common multiple)



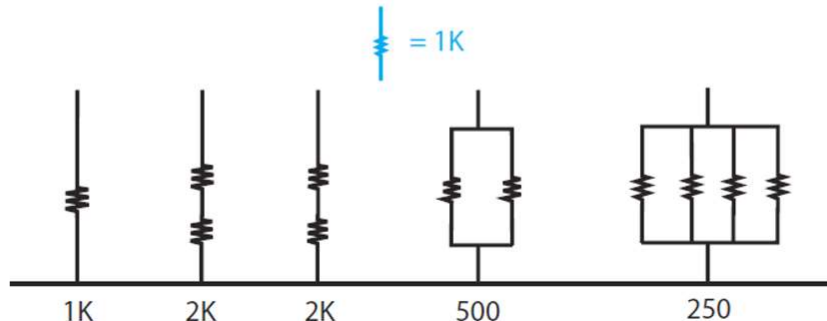
Matching: Unit Elements

❑ Terrible



❑ Better (use middle value)

- Less contact resistance (their variation is worse)
- Larger area (less mismatch) (overall area is roughly the same)
- Note: Do NOT connect unit caps/transistors in series

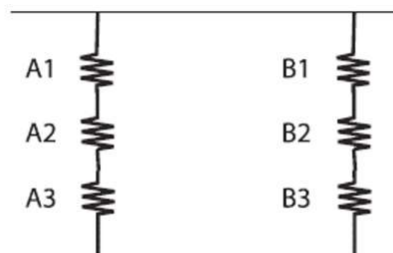


Matching: Interdigitation

- Place the root component (LSB) in the center



half of D	C	A root	B	half of D
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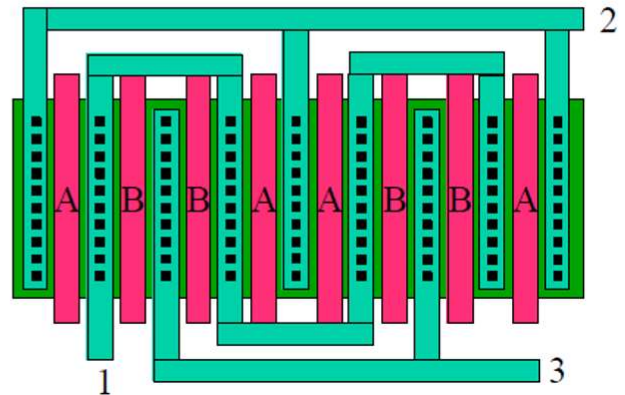
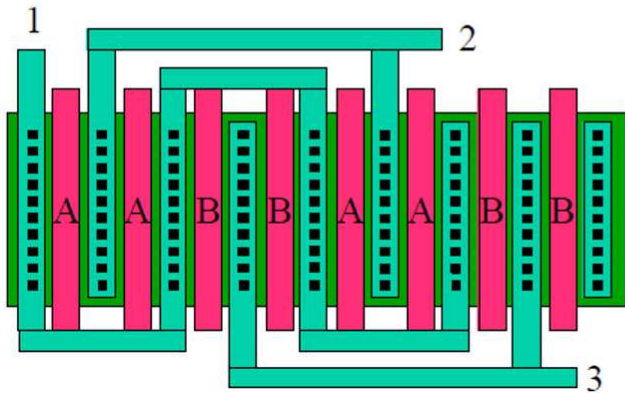
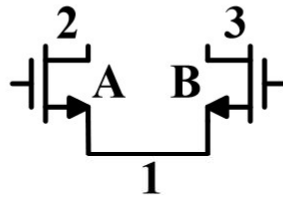
A1	B3	A2	B2	A3	B1
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Matching: Interdigitation

□ Each of A and B is 4 fingers

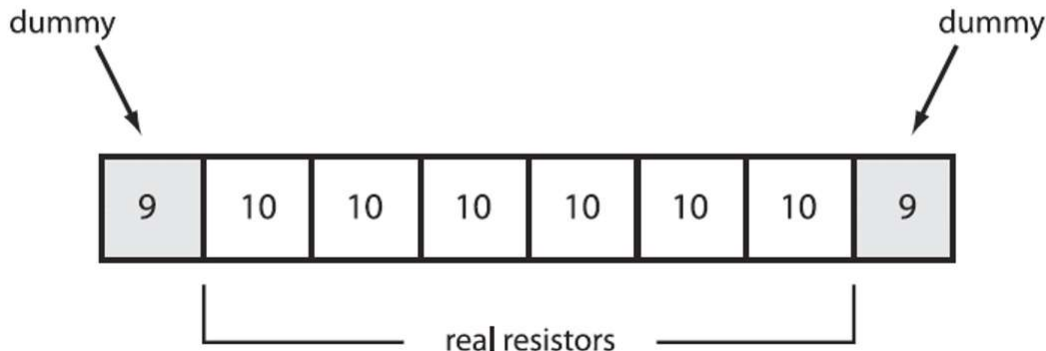
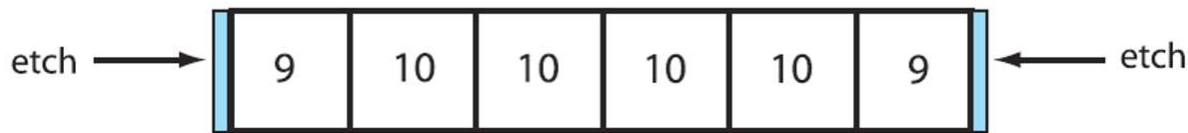
– AABBAABB

– ABBAABBA



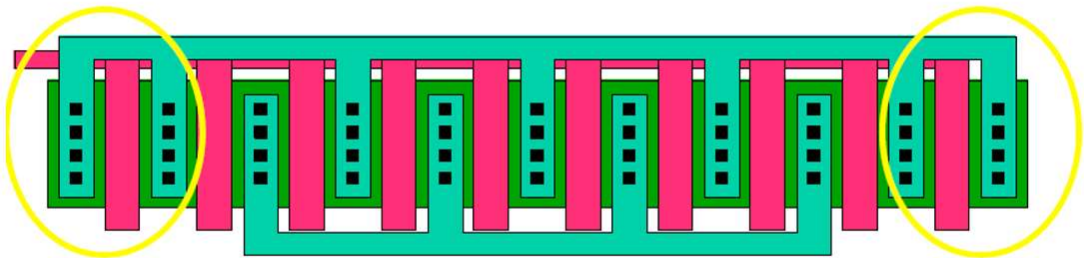
Matching: Dummies

- ❑ Structures at the ends have different boundary conditions compared to structures in the middle
- ❑ Edges suffer from a big difference



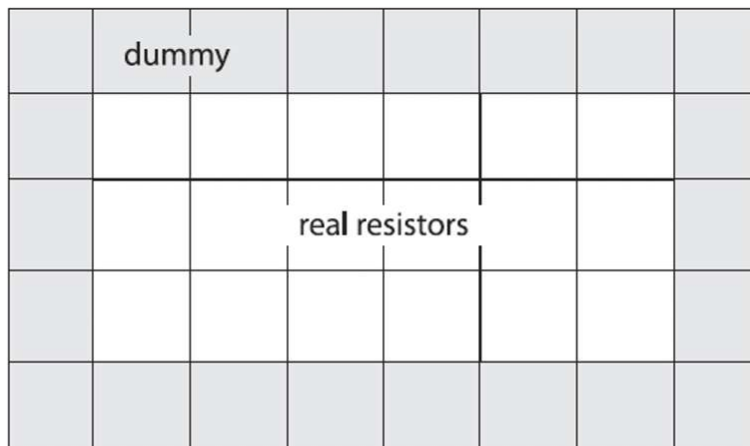
Matching: Dummies

- ☐ Dummy structures MUST be used
- ☐ Applies for transistors, resistors, and capacitors
- ☐ Dummy devices connected to GND (NMOS) or VDD (PMOS)
- ☐ Dummy gate should not be connected to input to avoid increasing capacitive loading



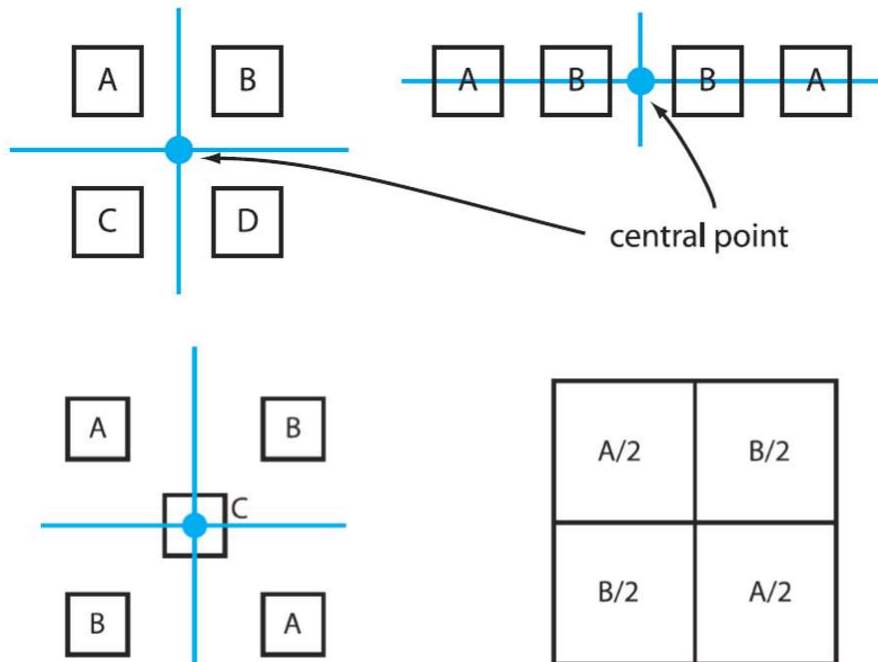
Matching: Dummies

- ❑ 2D arrays need dummies all around



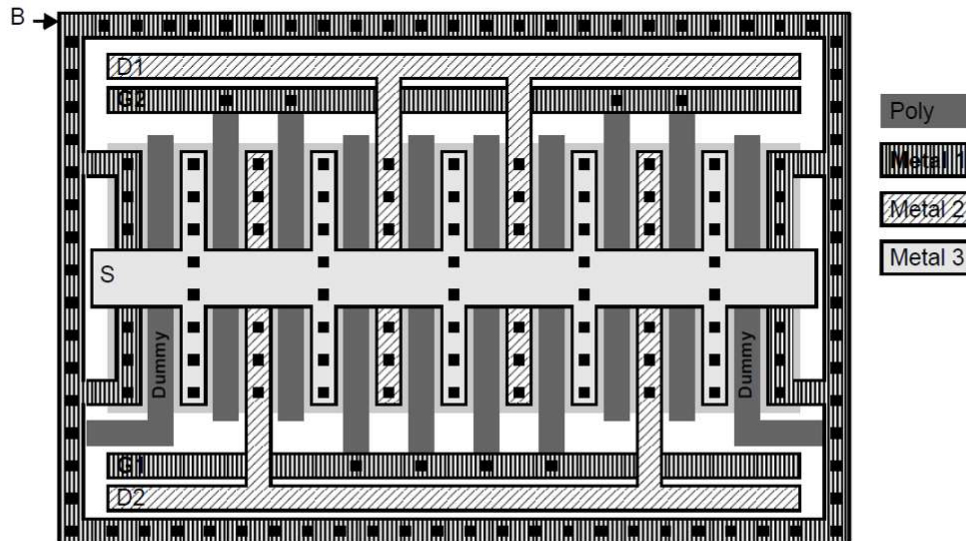
Matching: Common Centroid

- ❑ Reduces the effect of process and thermal gradients (1st order)
- ❑ Essential for differential input pair
- ❑ Cross-quad matches better than linear, but takes more time to do



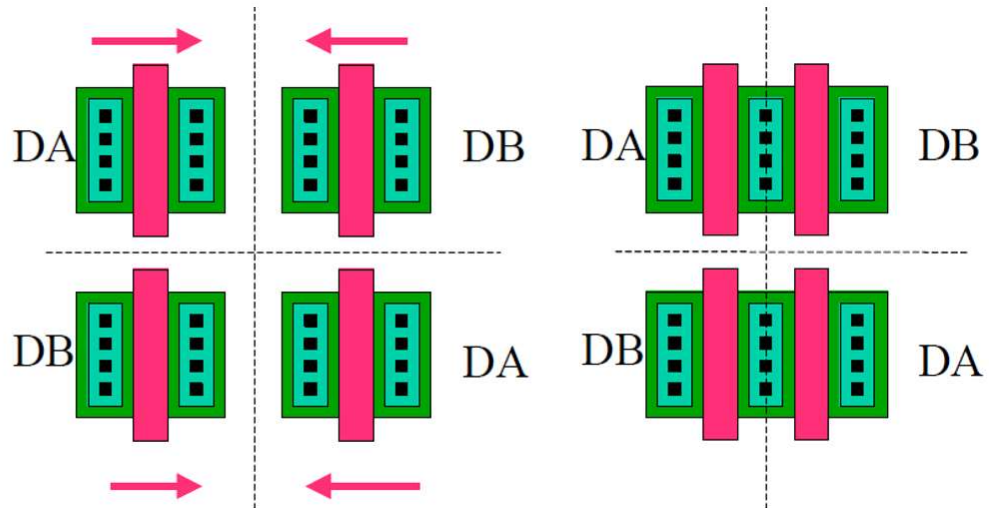
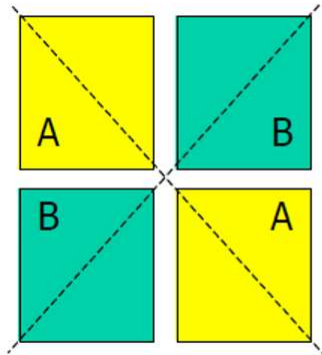
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- ❑ Ex: diff pair with AABBBBAA arrangement



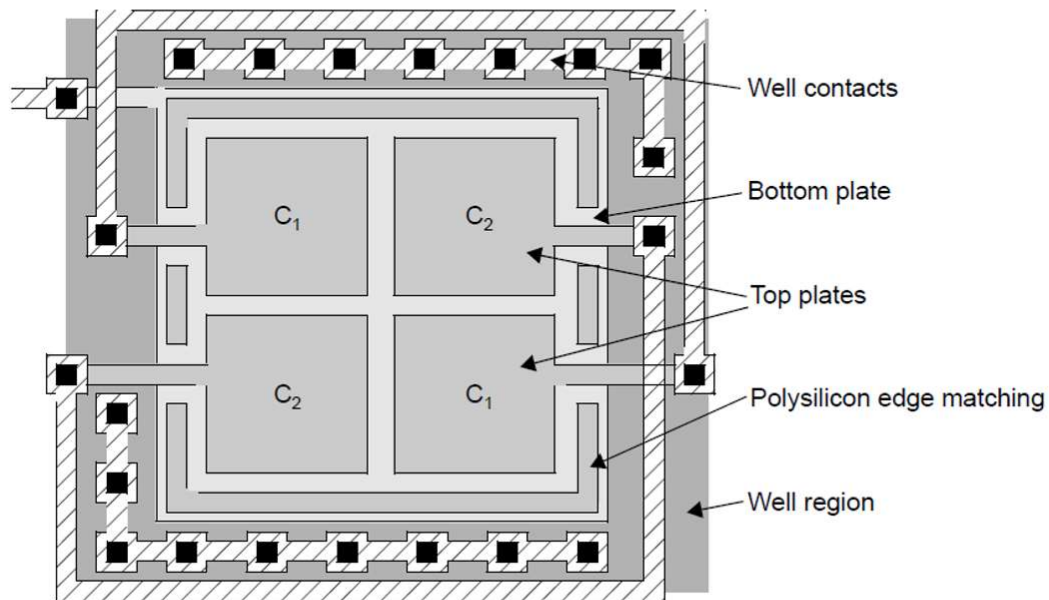
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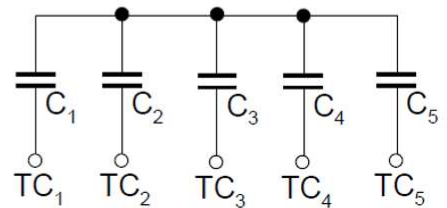
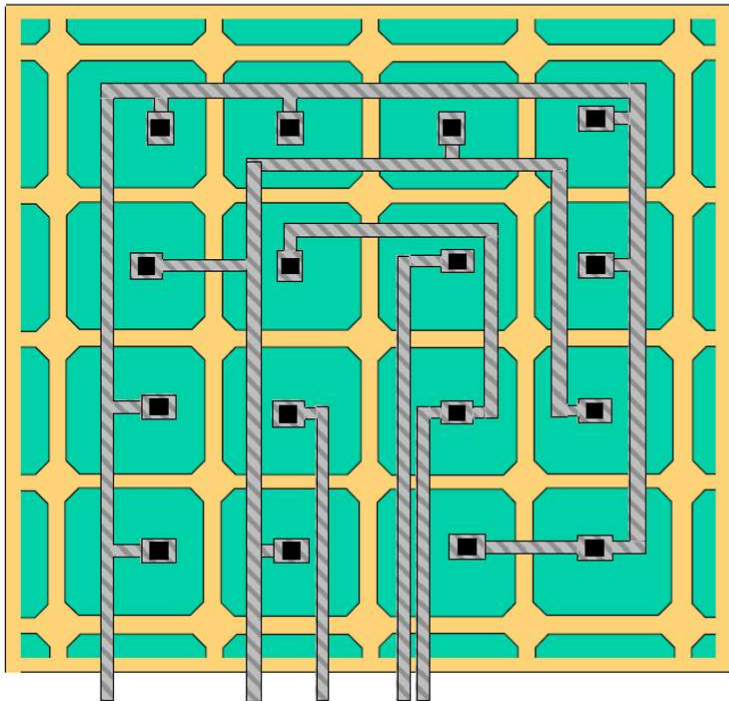
Matching: Common Centroid

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Matching: Common Centroid

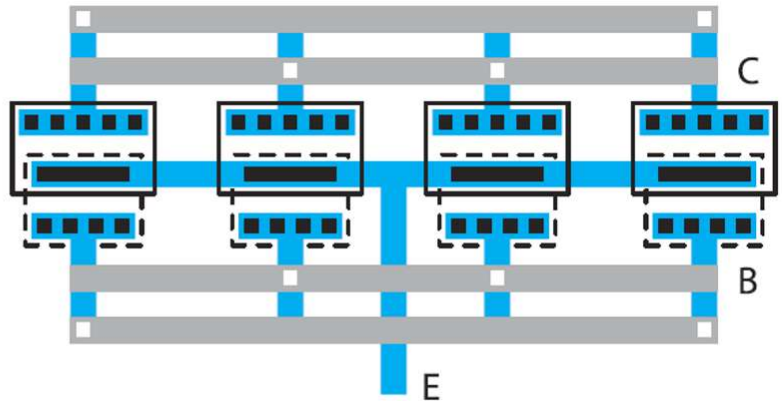
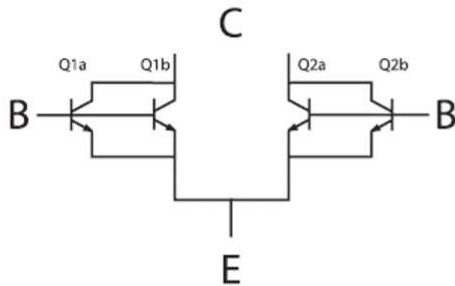
- ❑ Reduces the effect of process and thermal gradients



$$\begin{aligned} C_2 &= C_1 \\ C_3 &= 2C_1 \\ C_4 &= 4C_1 \\ C_5 &= 8C_1 \end{aligned}$$

Matching: Match the Parasitics

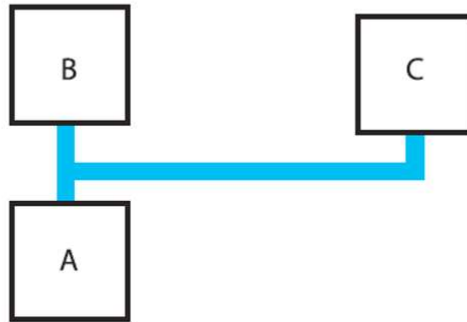
- ❑ Use extra metal overlaps to match wiring capacitance



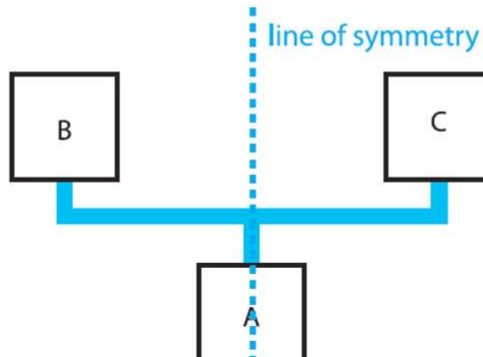
Matching: Symmetry

- ❑ A is supposed to talk to B and C in the same way
- ❑ Capacitance to B and C should be matched

❑ Bad



❑ Good



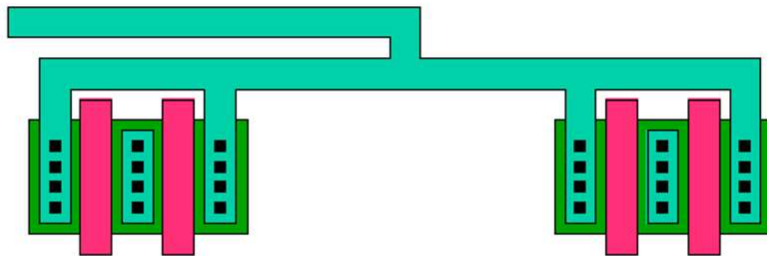
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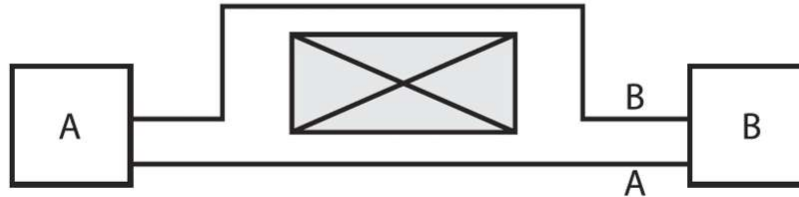


☐ Good

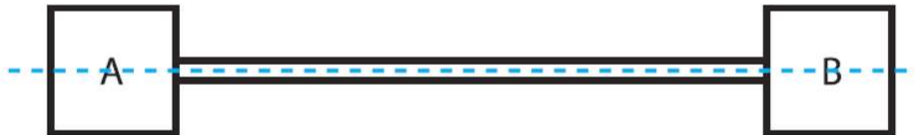


Matching: Differential Routing

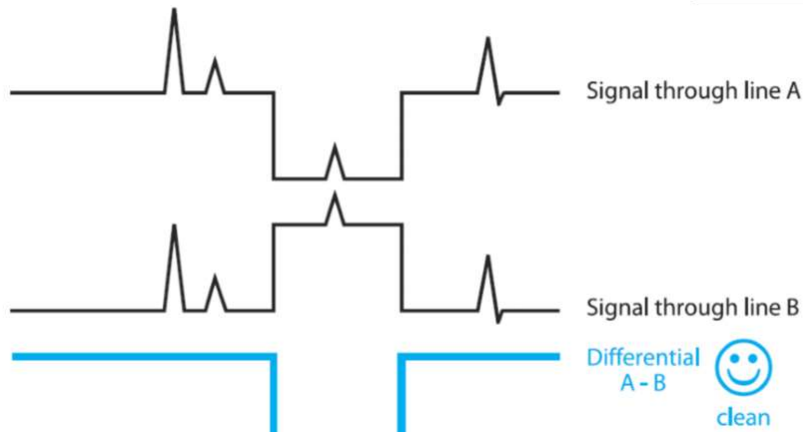
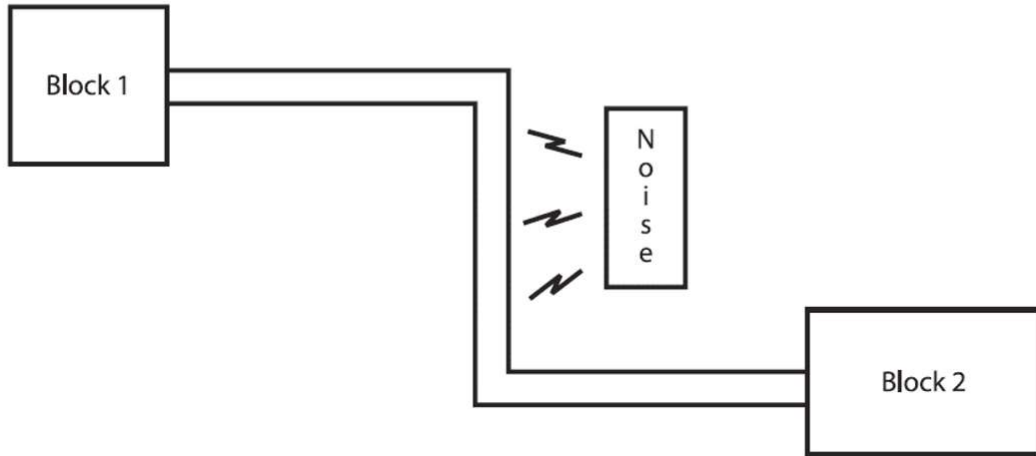
☐ Bad



☐ Good



Matching: Differential Routing



Matching: Go Large

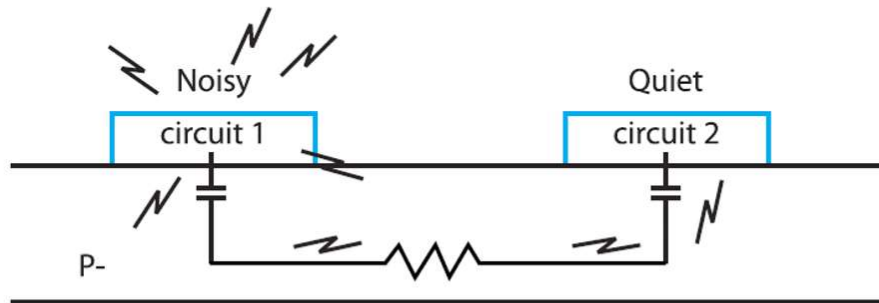
- ☐ Remember Pelgrom's model
- ☐ Never use min feature sizes

Matching: Summary

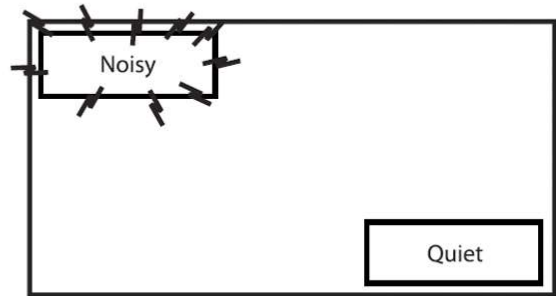
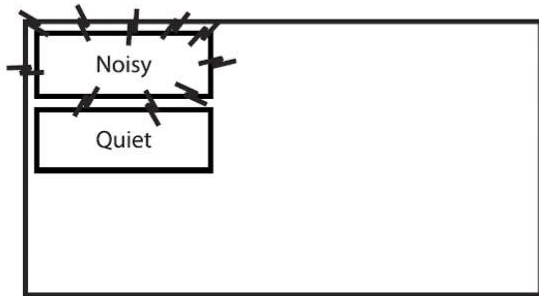
- Place matched devices close to each other.
- Keep devices in the same orientation.
- Choose a middle value for your root component.
- Interleave or interdigitate.
- Surround yourself with dummies.
- Cross-quad your device pairs.
- Match the parasitics on your wiring.
- Keep everything in symmetry.
- Make differential wiring identical.
- Match device widths.
- Go large.

Substrate Noise

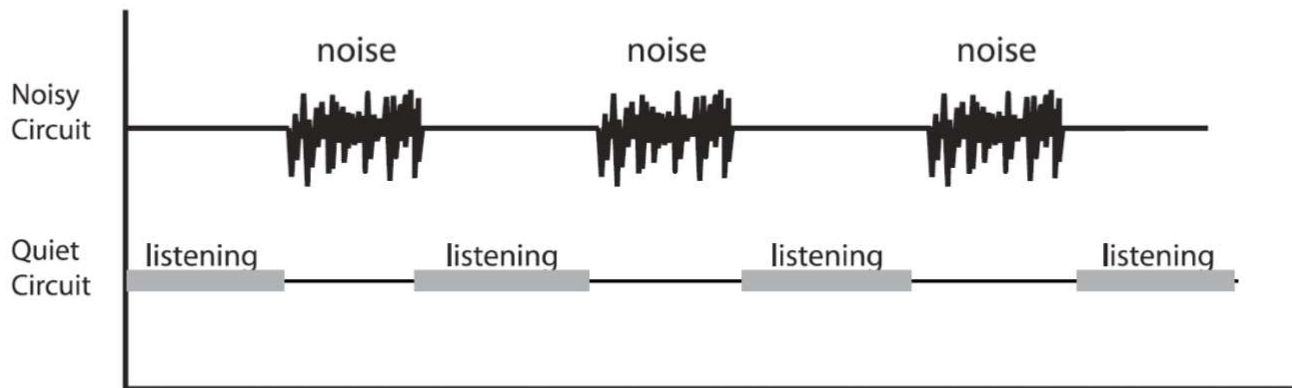
- ❑ Very critical issue in mixed-signal chips



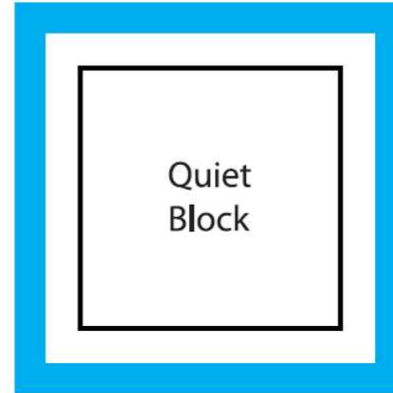
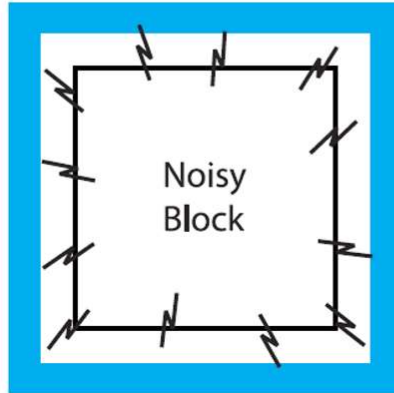
Substrate Noise: Go Far Away



Substrate Noise: Timing

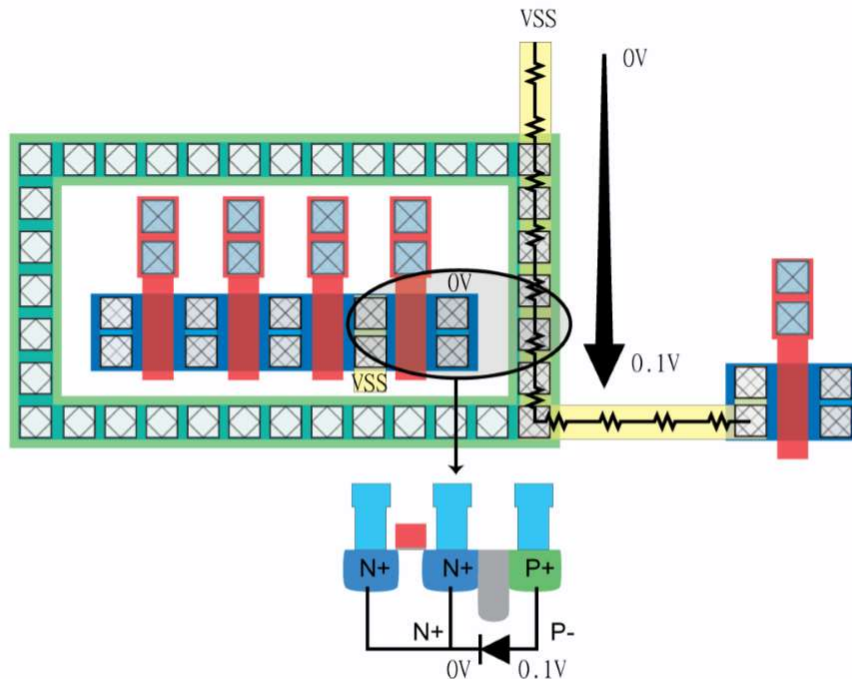


Substrate Noise: Guard Rings



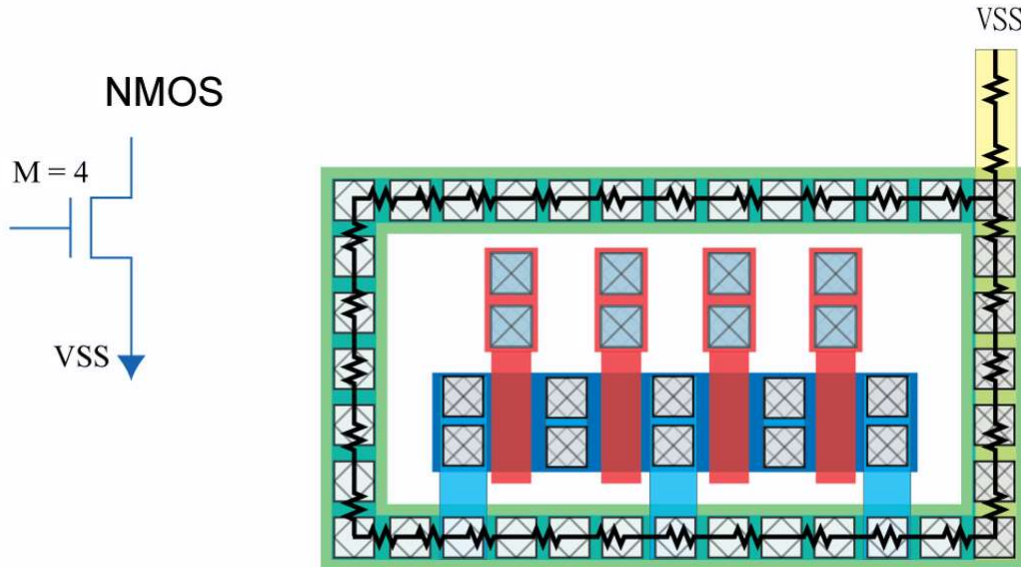
Substrate Noise: Guard Ring

- ❑ Avoid ANY current flow in guard rings
- ❑ Best practice: connect the substrate and guards with a separate star connection to the GND pad (or use double bonds)



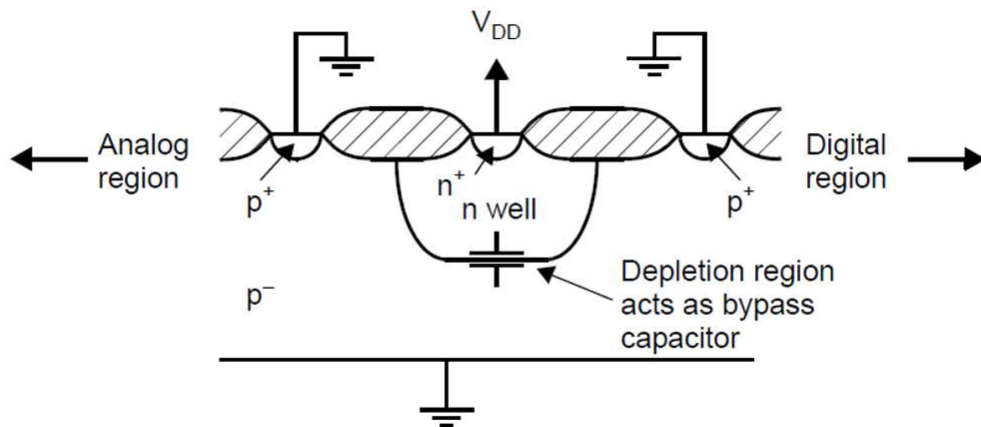
Substrate Noise: Guard Ring

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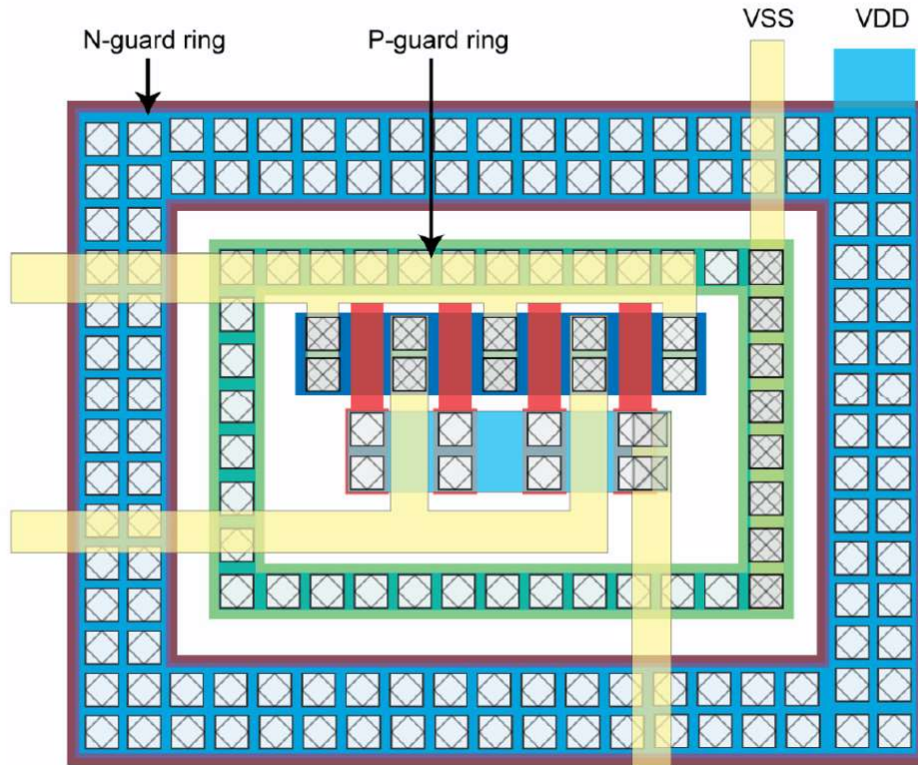
Substrate Noise: N-Well Barrier

- ❑ Separate analog and digital areas by an n-well
 - High resistance path between analog and digital substrates
 - Minimize the injection of noise from digital circuits into the substrate under the analog circuit
- ❑ The well also acts as a bypass capacitor



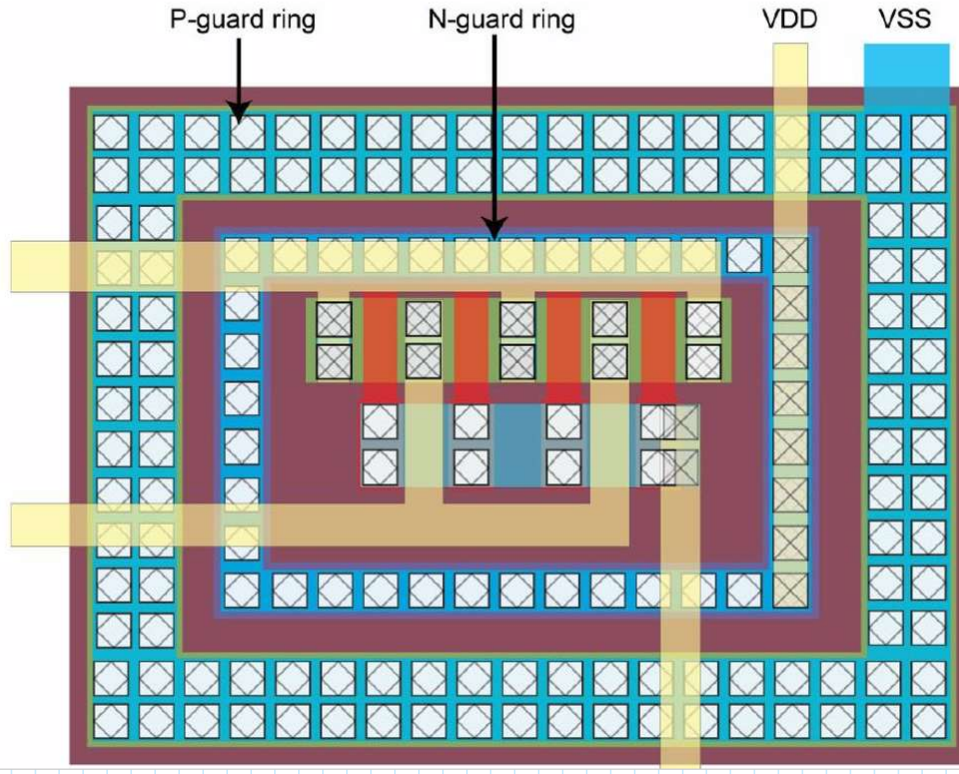
Substrate Noise: Double Guard Ring

- ❑ Conservative layout: VDD guard ring collect stray electrons in the substrate

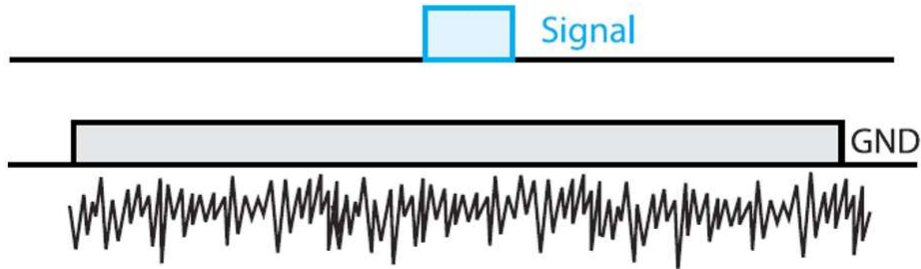


Substrate Noise: Double Guard Ring

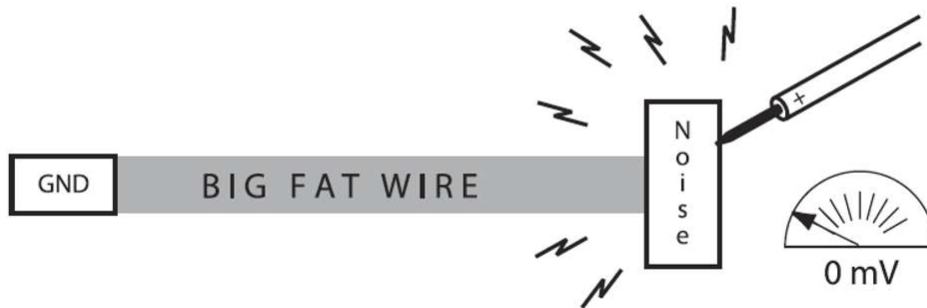
- ❑ Conservative layout: GND guard ring collect stray holes in the substrate



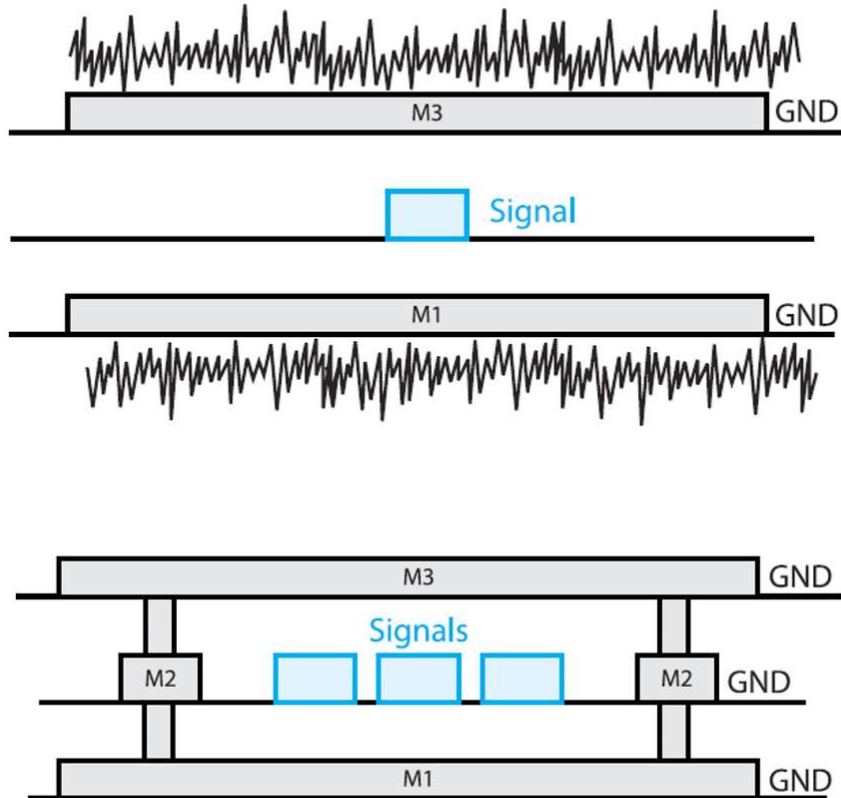
Substrate Noise: Shielding



- ❑ The GND line used for shielding must have quite low resistance

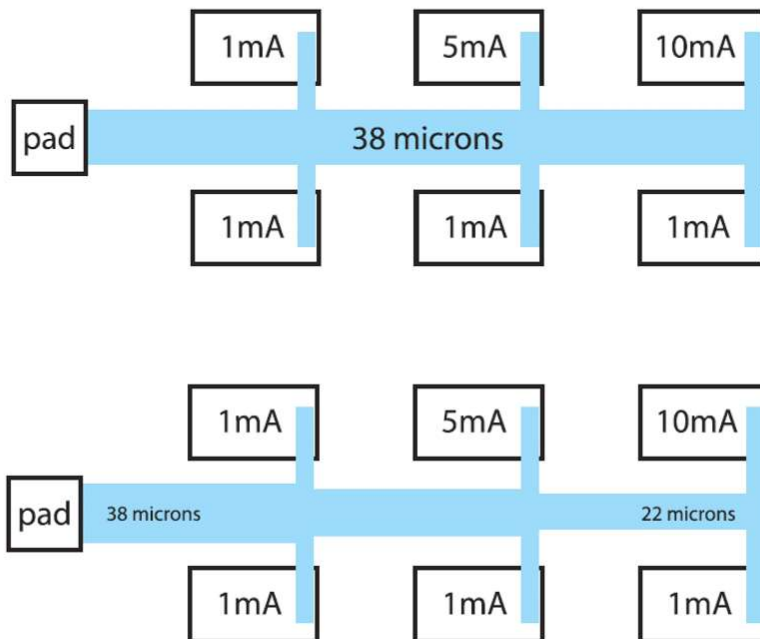


Shielding Sensitive/Noisy Signals



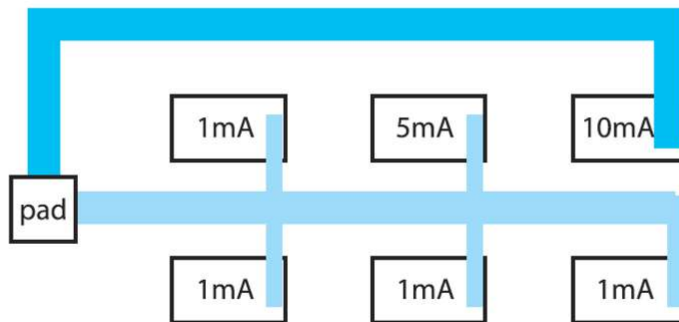
Routing Power Rails

- ❑ Better to place high current blocks near power pad
- ❑ Wire width can be tapered



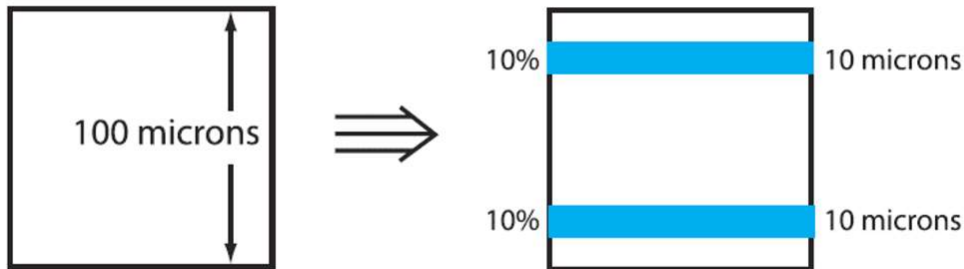
Routing Power Rails

- ❑ Better to place high current blocks near power pad
- ❑ Wire width can be tapered
- ❑ Better technique: high current power line should be routed independently
 - Star vs bus



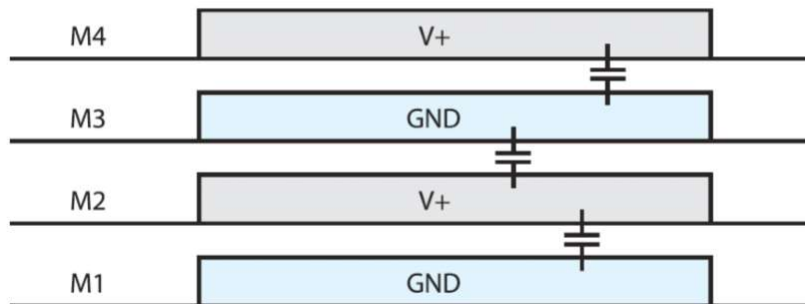
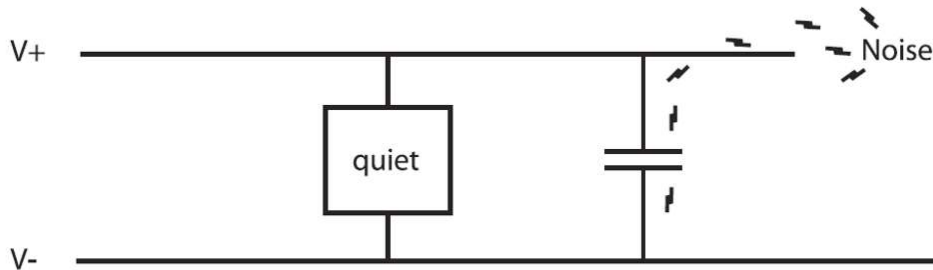
Routing Power Rails

- ❑ Use lowest resistance metal layer
- ❑ Use FAT wires
 - Do not use min width
 - Do not stress the wire at its max current density
 - Roughly 10% of the cell height



Supply Decoupling

- ❑ Decoupling provide low resistance path for high frequency noise
- ❑ Fill any white space in the layout with decoupling caps
- ❑ Stacked power rails give some free decoupling caps

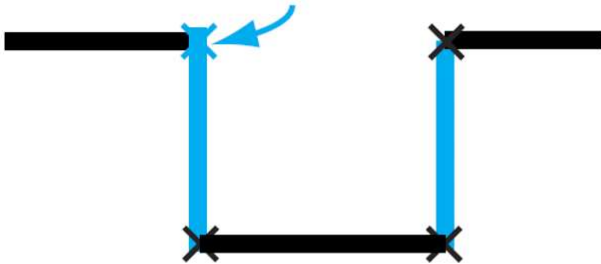


Vias

- ❑ For analog and high current signals
 - Do NOT use min metal width
 - Do NOT use single via

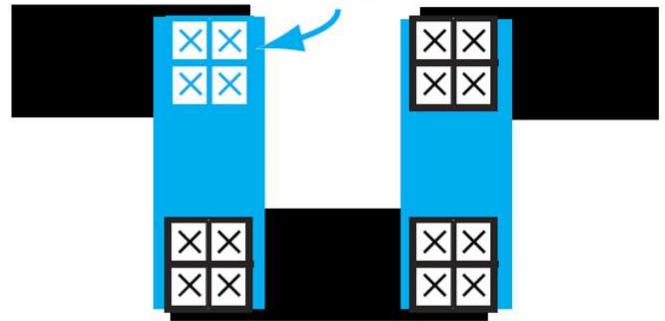
Bad

single via



Good

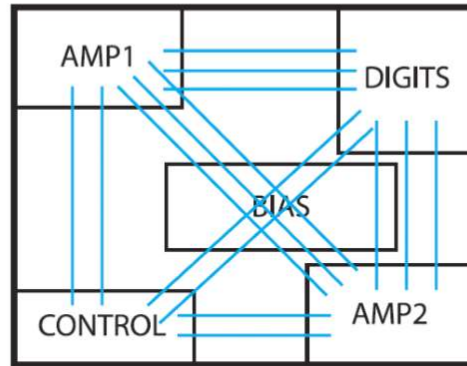
four vias



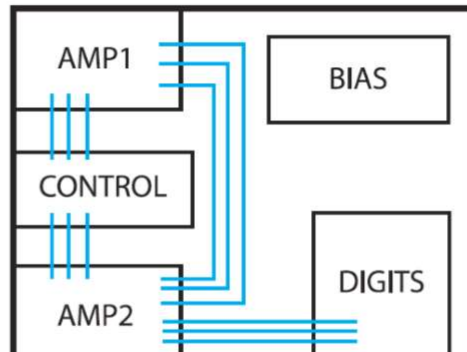
Floorplanning: Wiring

☐ Neat flylines indicate good floorplanning

☐ Bad

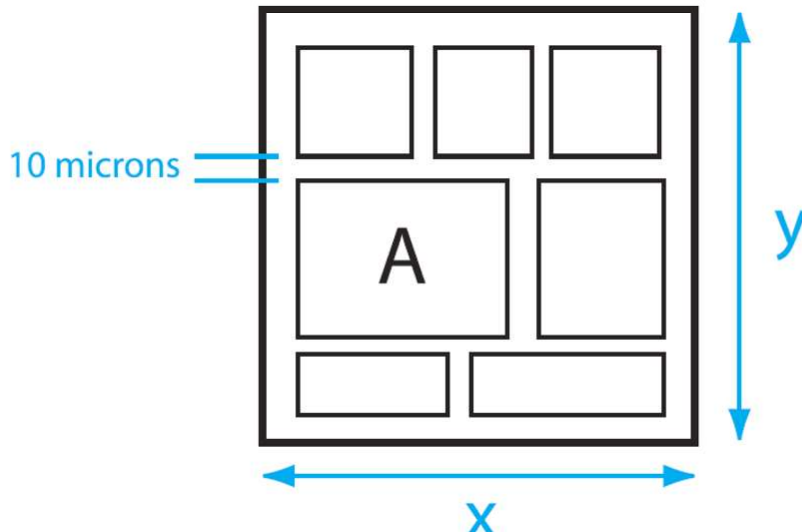


☐ Good



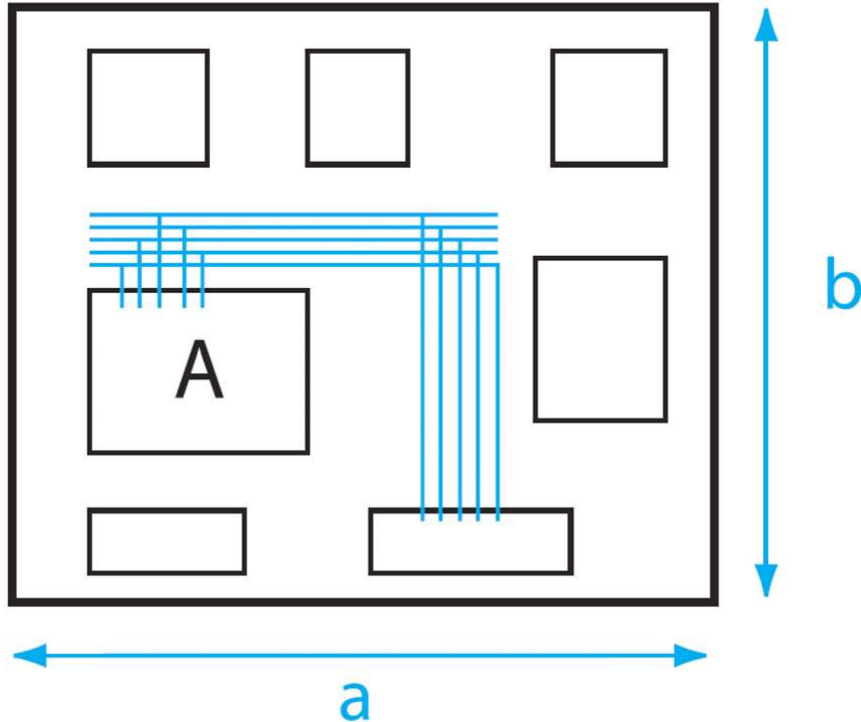
Floorplanning: Wiring

- ❑ Minimizing the area is good
 - But you must leave enough space for wiring
- ❑ Remember that “in general” you cannot route over other blocks
- ❑ Shielding, differential routing, power and clock routing, guard rings, etc., all take a lot of area



Floorplanning: Wiring

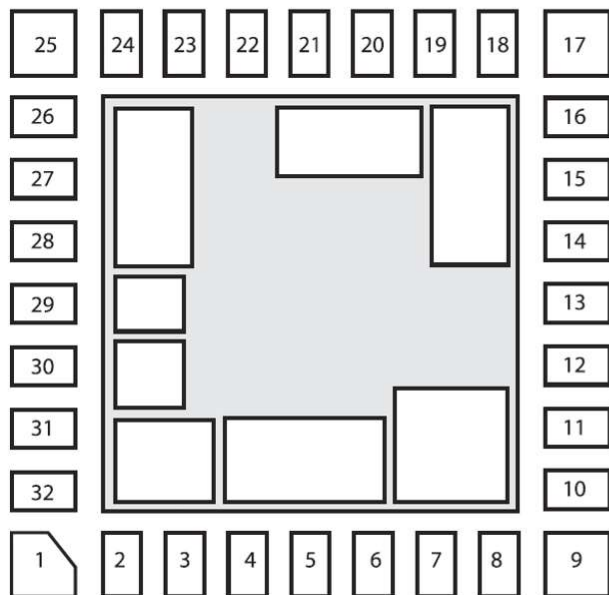
- ❑ Bigger area, but more realistic



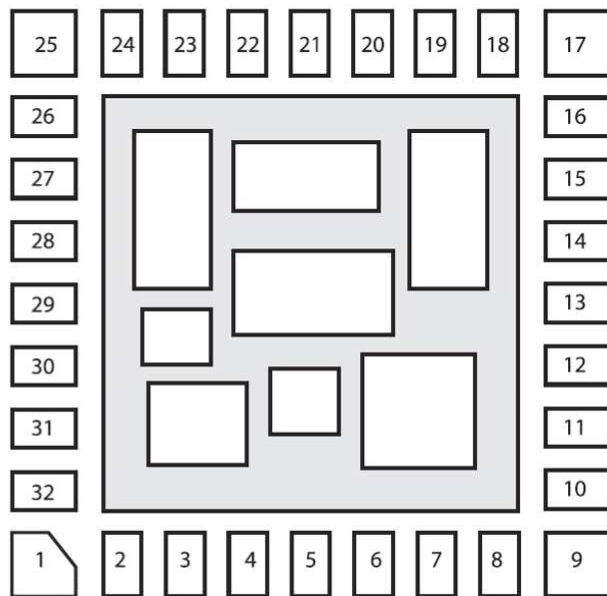
Floorplanning: Wiring

- ❑ Leave some free space for wiring, decoupling caps, last minute updates, etc.

Bad

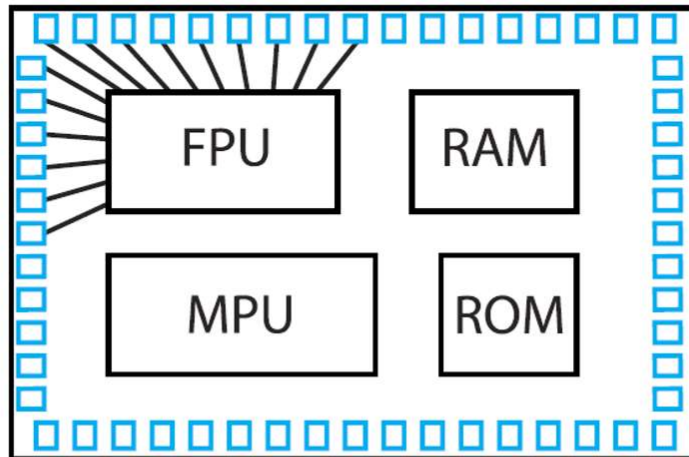


Good



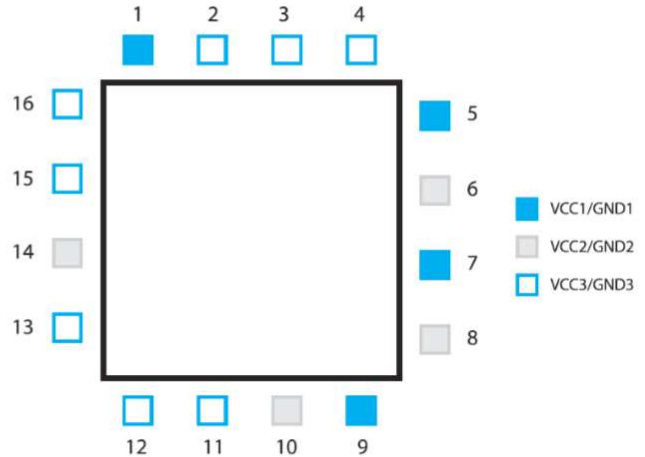
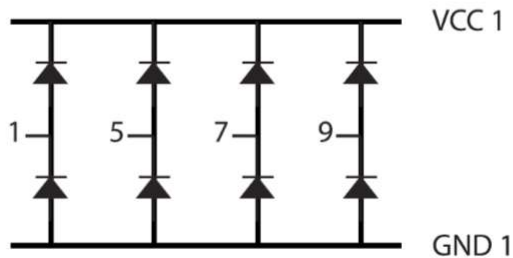
Floorplanning: Pinout

- Inputs and outputs should be located near their appropriate cell block

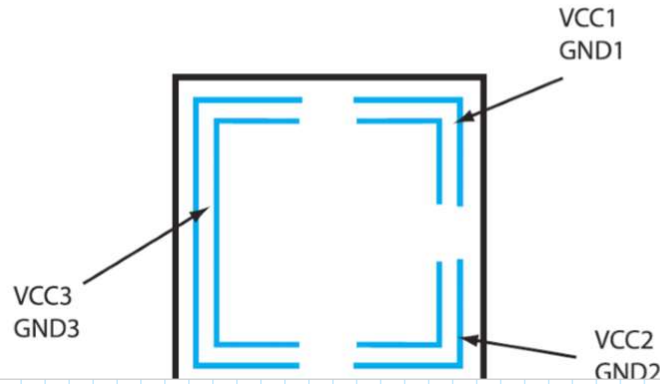


Floorplanning: ESD

☐ Bad

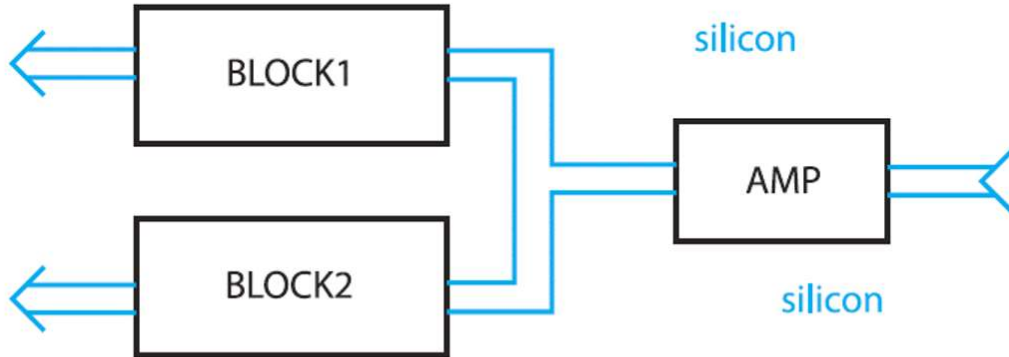


☐ Good



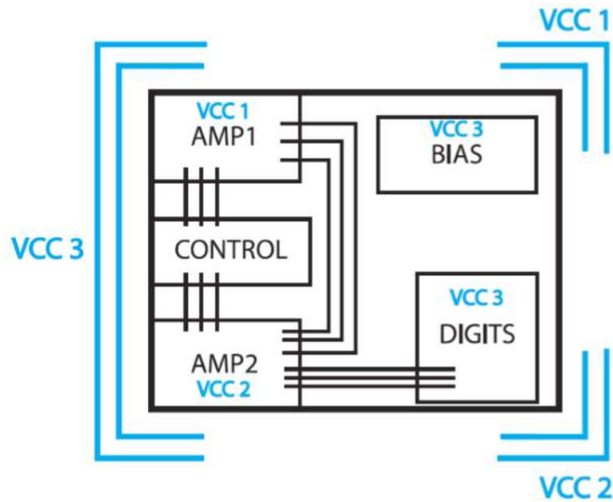
Floorplanning: Signal Flow

- ❑ Flow of sensitive signals may dictate specific block placement
 - Specifically important for RF signals at very high frequencies

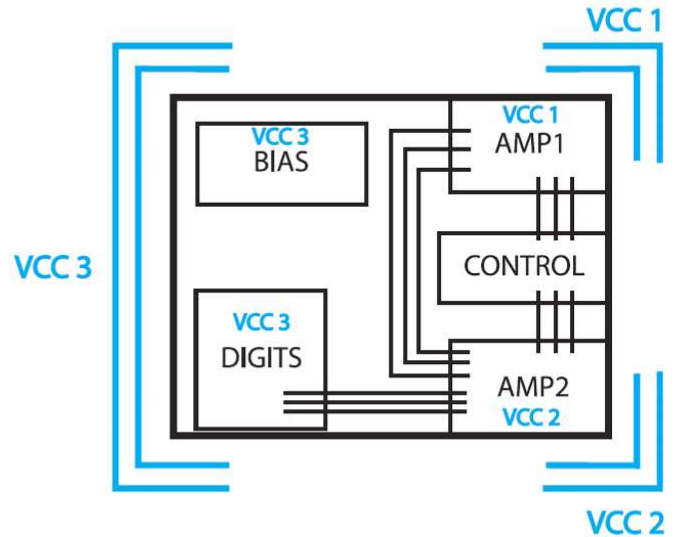


Iterative Floorplanning

Bad

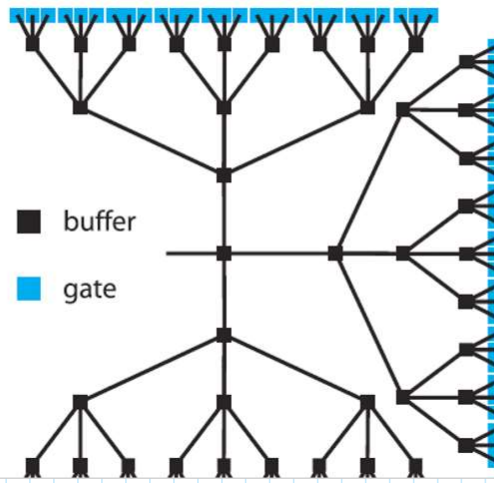
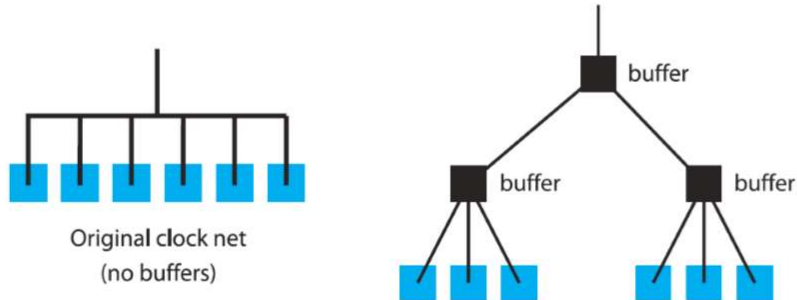


Good



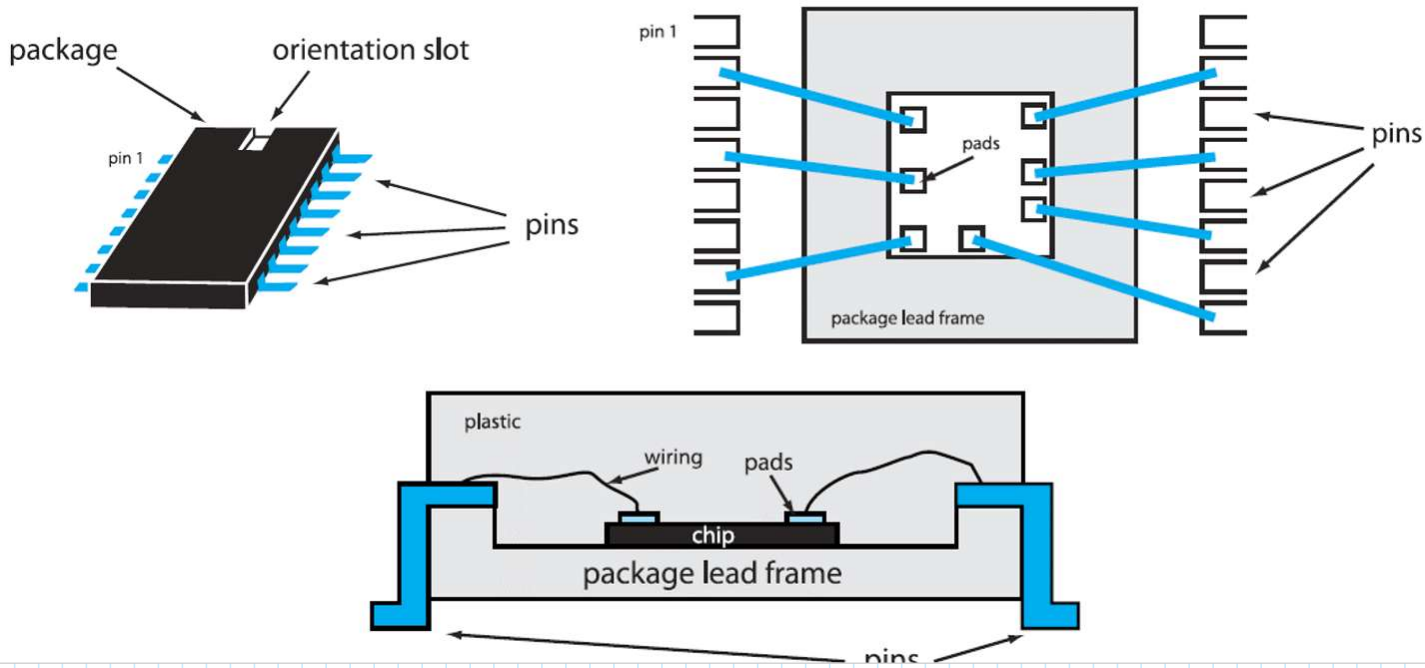
Clock Tree

- ❑ The clock should tick at all elements exactly at the same time



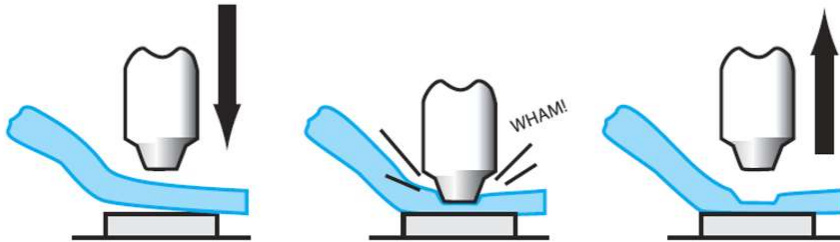
The Package

- ❑ The package type must be planned even before floorplanning
 - The pinout (floorplanning)
 - Chip area (the chip must fit inside the package)

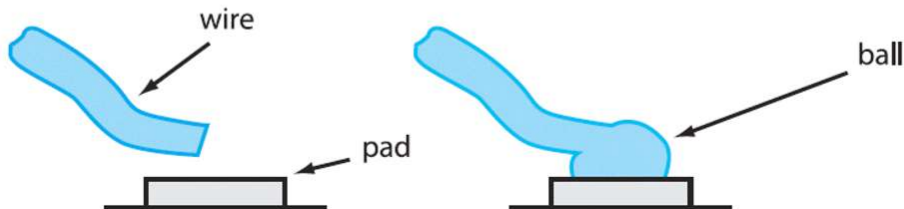


Wire Bonding

- ❑ The bond wire is ultra-thin ($\sim 25\mu\text{m}$) Gold or Aluminum wire
- ❑ Ultrasonic wedge bonding
 - Uses brute force (pressure) and ultrasound (vibration)



- ❑ Ultrasonic ball bonding
 - Uses high voltage and ultrasonic to melt the wire



Additional Topics

❑ Design and layout in FD-SOI 22nm technology

- <https://www.globalfoundries.com/resources/technical-webinar-series/analog-design-workshop-22fdx-22nm-fd-soi-technology-part-one>
- <https://www.globalfoundries.com/resources/technical-webinar-series/analog-design-workshop-22fdx-22nm-fd-soi-technology-part-two>
- <https://www.globalfoundries.com/resources/technical-webinar-series/top-5-design-guidelines-successfully-implement-22fdx-fd-soi-technology>

Thank you!